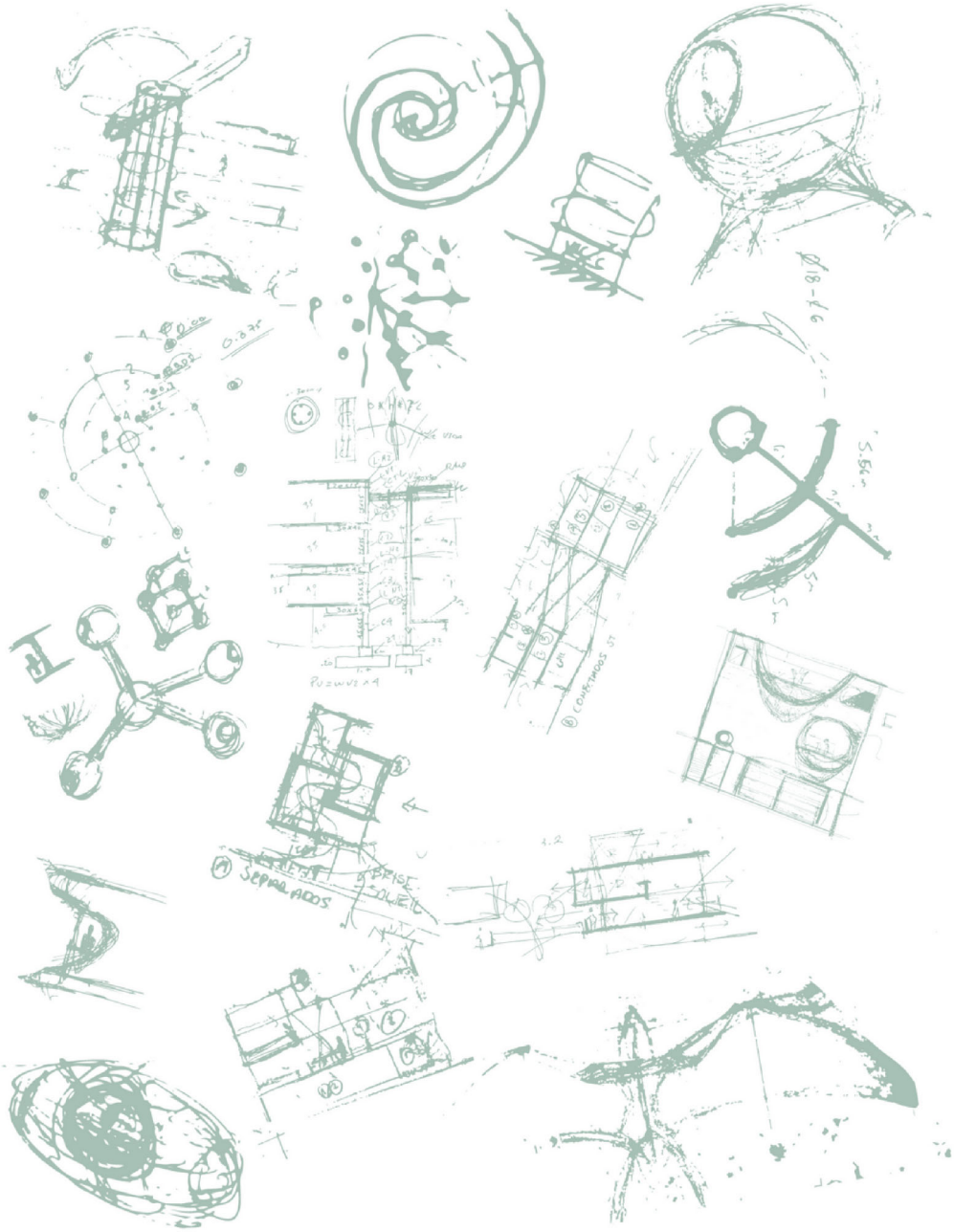
portfolio.

Selection of Architectural Works + CV

PABLO DEL ANGEL

2024



Pablo Andrés Del Angel Zambrano

28/02/01

Bachelor of Architecture at Tecnológico de Monterrey

Monterrey, Nuevo León, MX

CONTACT INFO

Email
pablodelangel00@gmail.com

Phone
(+52) 8122045988

Photography
Instagram
@delangel.pa

Languages

Spanish	English
Portuguese	French
Catalan	Hindi

Education

Monterrey, N.L - Expected Graduation 2025
School of Architecture, Art and Design EAAD
Insitituto Tecnológico y de Estudios Superiores de Monterrey

Cultural Development

Monterrey, N.L - 2007-2016
Escuela Superior de Musica y Danza de Monterrey (ESDMD)
Music studies, for instruments piano and percussions.

Monterrey, N.L - 2016-Present
Orquesta Sinfonica del Tecnológico de Monterrey
Orchestral Percussionist & Section Leader.

Monterrey, N.L - 2016-Present
Scholarship for Musical Talent in Tecnológico de Monterrey

Awards

Proyecta Awards ED 22 2023
*Nominated by **ITESM EAAD** in the Community Equipment Category with "**Project Paws**".*

Proyecta Awards ED 23 2023
*Nominated by **ITESM EAAD** in the Collective Housing Category with "**RYUKO-TO**".*

Experience

Monterrey, N.L. - 2021
Renderer at Dos Islas Decor (Interior Design)

Monterrey, N.L - 2023
Construction Control at Constructora Sepulveda del Norte

Social Service

Monterrey, N.L - 2023
Construction work for public school. With *Impulso Urbano*

Leadership Development

Mumbai. India 2018-2019
Mexican Ambassador with Rotary Youth Exchange Program (Cultural Exchange)
Rotary Club D4130 → D3142

Tenerife, Canary Islands (remote) 2022 - Present
Image Committee President
Rotaract La Laguna D2201

Monterrey, N.L 2022 - 2023
Club President
Rotex Club D4130

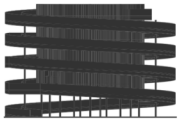
Software Skills

Autodesk Revit	Adobe Photoshop	Unreal Engine
Autodesk Autocad	Adobe Illustrator	QGIS
Autodesk 3Ds Max	Adobe Lightroom	**Digital Fabrication**
Twinmotion	Adobe Premiere	3D printing
Rhino + Grasshopper	Adobe InDesign	Laser Cutting

content.

Obispado Station
Opus 01

Ryuko-To
Opus 02



Helix Vitae
Opus 03

Plaza Dino
Opus 04

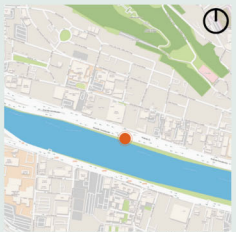
RT Pavilion
Opus 05

Doppio
Opus 06

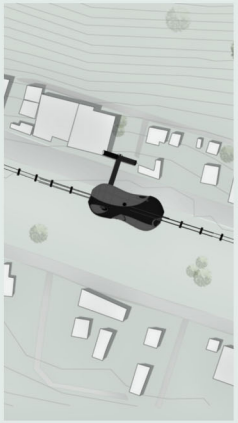
Oculus
Opus 07

Dome Shelter
Opus 08

Tensile Shade
Opus 09



Constitución 930, Real del Valle 1 Sector, 64060 Monterrey, N.L.



Walkthrough Video

Colaborators:
Isabela Cárcoba
Valeria López

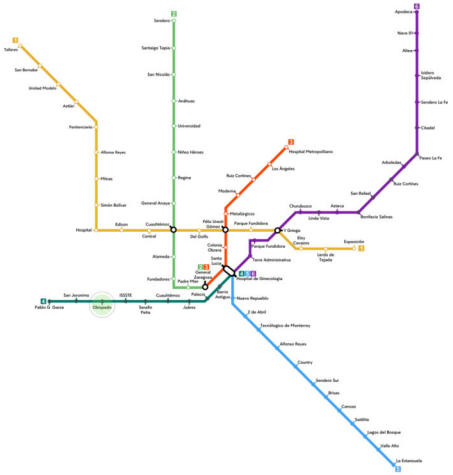
2024

Obispado Station

"An urban icon where transportation, design, and functionality intersect.."

In the context of Monterrey, Mexico's growing urban development, designing a metro station that enhances urban mobility while integrating harmoniously with its surroundings is a significant challenge. The Obispado station, part of the new Line 4 of the metro network, will be situated in front of the iconic Obispado Towers and the future Torre Rise, the tallest building in Latin America. Located on Avenida Constitución with the Río Santa Catarina to the south, this project aims to optimize user experience by highlighting significant views, improving wind flow, and effectively managing solar radiation.

The Metrorrey system serves as Monterrey's metro system, enhancing urban mobility with its interconnected lines. Line 4, specifically, aims to enhance connectivity from Miravalle to the junction with lines 5 and 6, spanning 36 kilometers and encompassing several significant city areas.

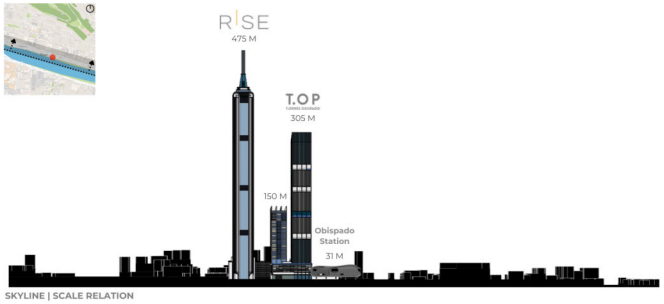


Existing Lines

1 Talleres ↔ Exposición
2 Sendero ↔ General Zaragoza
3 Hospital Metropolitano ↔ General Zaragoza

New Lines

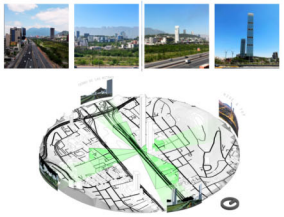
4 Pablo G. Garza ↔ Hospital de Ginecologia
5 La Estanzuela ↔ Hospital de Ginecologia
6 Apodaca ↔ Hospital de Ginecologia



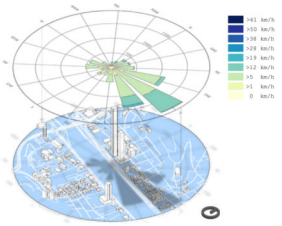
SKYLINE | SCALE RELATION

The site analysis reveals prevailing east winds and varying seasonal speeds, affecting the general design of the station. Solar exposure presents a significant challenge, requiring solutions for radiation management and interior comfort.

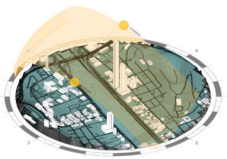
The project adopts an approach that highlights significant views such as the Museo del Obispado, the Cerro de la Silla, the Sierra Madre, the Ave building, and the TOP and Rise towers. This visual concept is complemented by other formal considerations like wind and solar exposure.



VIEWS
To highlight visual integration with the surrounding landscape.



WIND DIRECTION
To reduce wind resistance and structural pressure.



SUNPATH
For thermal regulation and radiation impact reduction.

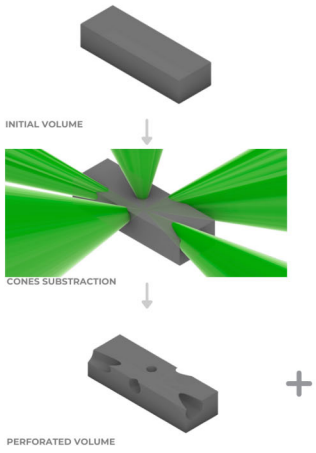
Design Considerations

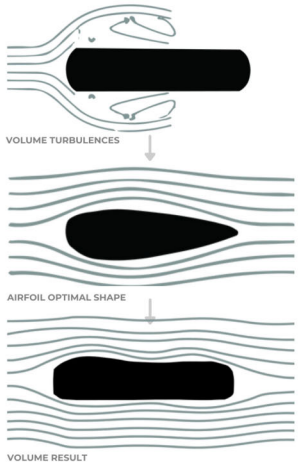
Visual Connection:
The design begins by identifying strategic viewpoints from the station's center to the highlighted visual elements. These points are connected by lines, which, when considering viewing angles between 20 and 120 degrees depending on distance, transform into conical volumes, materializing the field of view. These volumes define the station's fields of view, starting with a rectangular prism to which subtractions based on the fields of view are applied, resulting in a perforated volume.

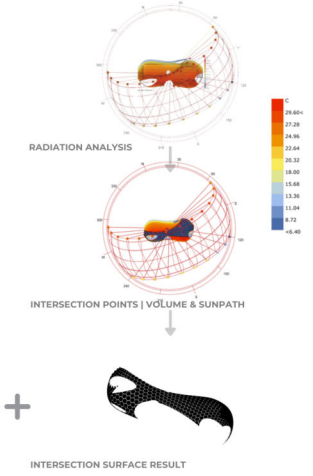
Wind Flow Optimization:
The initial prism created turbulence, so it was modified to adopt a cylindrical shape, thus improving wind flow. Further simulations and aerodynamic studies, considering concepts like laminar and turbulent flow, pressure points, and vortex generators, led to a final shape considering other aerodynamic base shapes and profiles like the airfoil, bullets and other optimal flow shapes. This form optimizes wind flow and minimizes turbulence without compromising the interior space.

Solar Radiation Management:
The south facade, exposed to the sun, required an effective shading solution. An analysis of the annual solar path identified the points of highest radiation. The solution was to design a second layer as a shade based on these intersection points, forming a surface that covers the main volume when the sun directly impacts it. Additionally, dual-axis solar tracking panels were incorporated, which adjust their position to provide shade when necessary and allow natural light to enter at other times.

Formal Process & Design Guidelines







The project achieves a harmonious integration of the three key factors: highlighting significant views, optimizing wind flow, and managing solar radiation. The project addresses the site's challenges and needs, proposing solutions that enhance user quality of life and contribute to the city's urban development.

Kinetic Panels

Designed to adapt their orientation according to solar radiation for optimal natural lighting and heat reduction, enhancing energy efficiency and interior thermal comfort.

Main Shell & Skydome

Aluminum panels compliment with the glass skylight enabling natural light to enter, fostering a luminous environment, while regulating the temperature.

Secondary Structure

Steel profiles integrate with the primary structure, ensuring longevity and strength, while also enabling flexibility in architectural design.

Main Structure

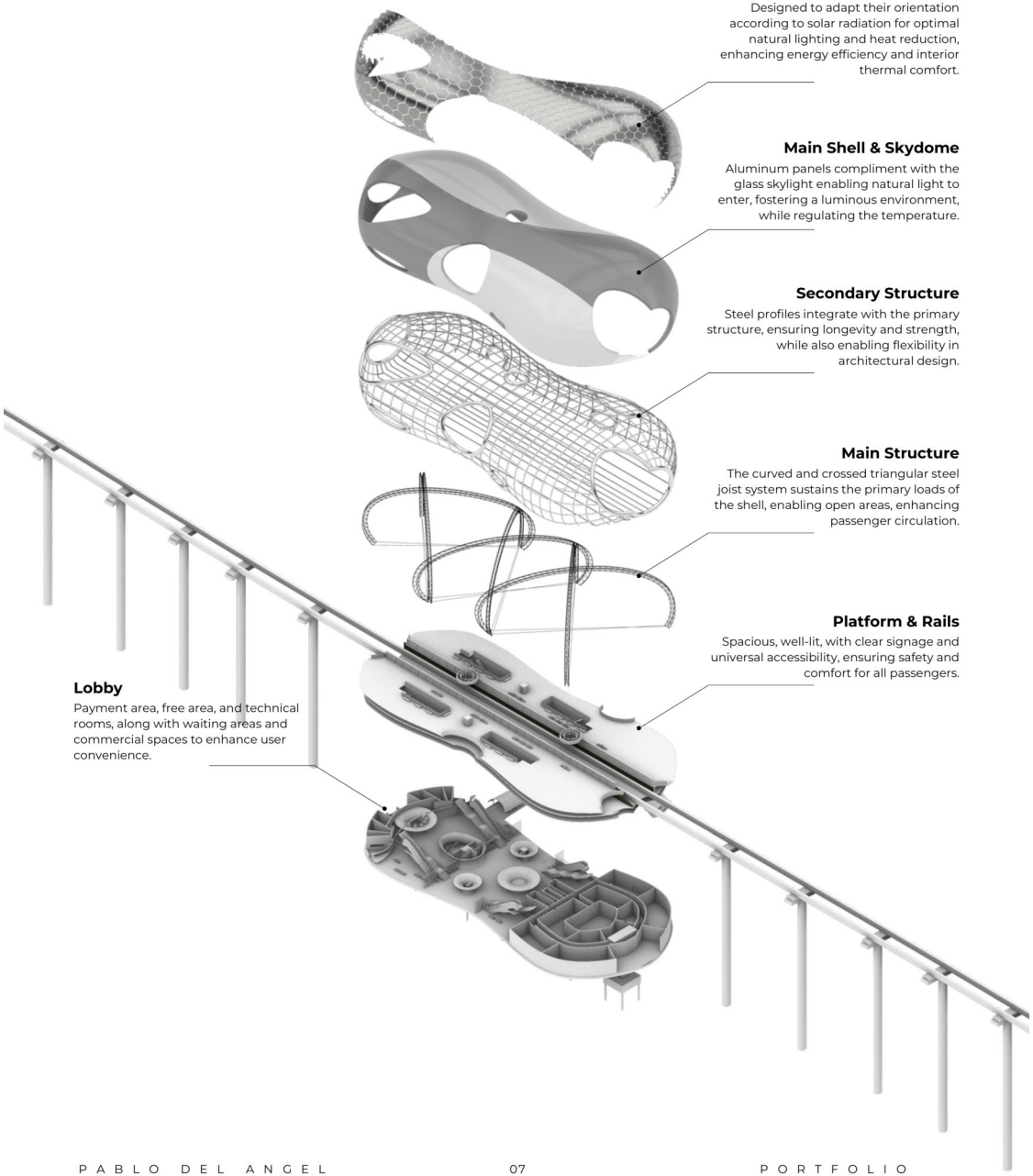
The curved and crossed triangular steel joist system sustains the primary loads of the shell, enabling open areas, enhancing passenger circulation.

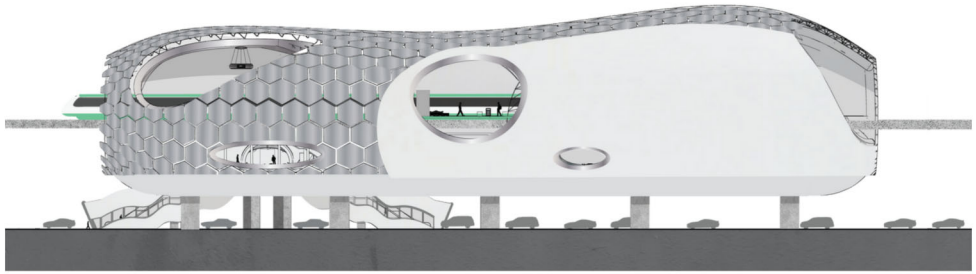
Platform & Rails

Spacious, well-lit, with clear signage and universal accessibility, ensuring safety and comfort for all passengers.

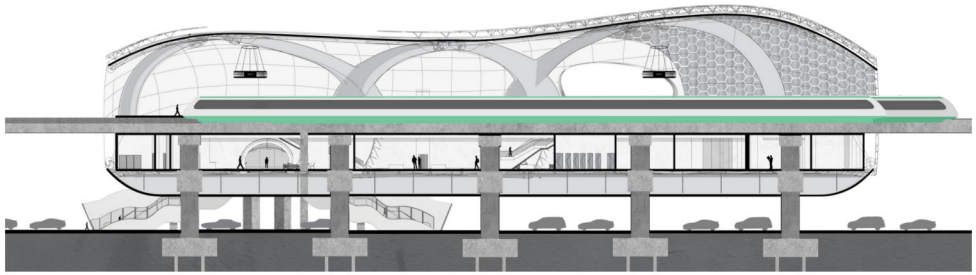
Lobby

Payment area, free area, and technical rooms, along with waiting areas and commercial spaces to enhance user convenience.

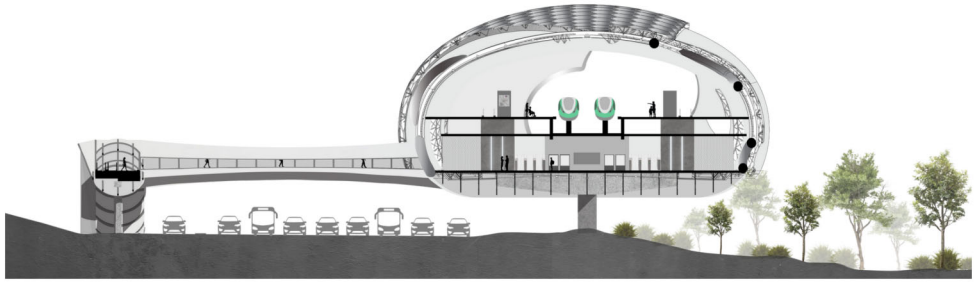




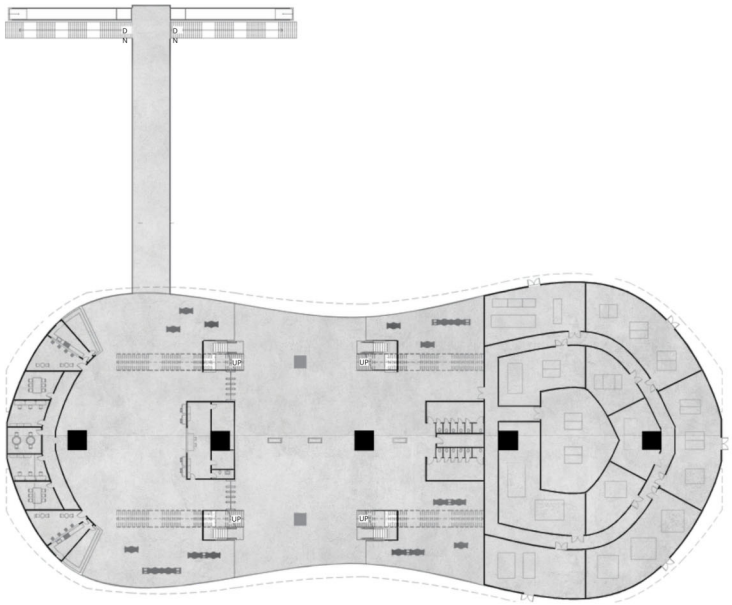
South Facade



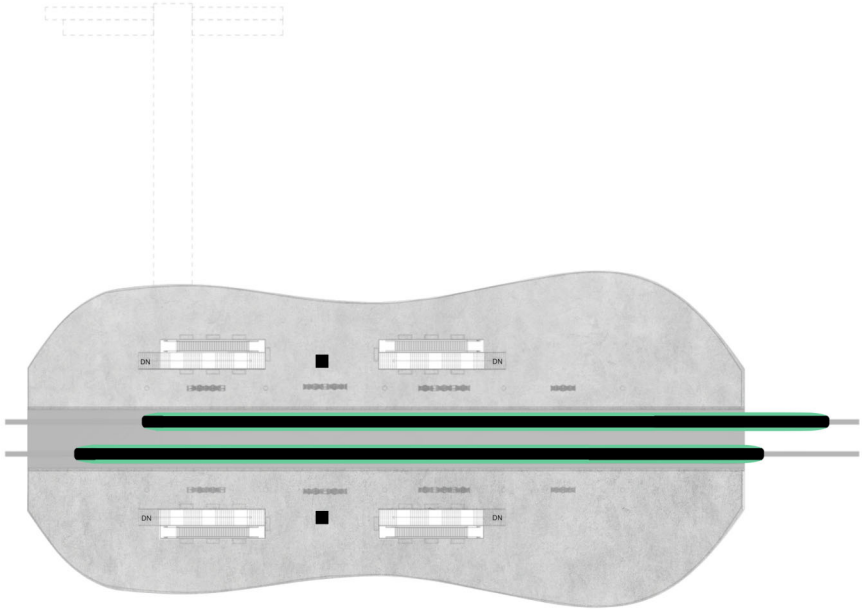
Longitudinal Section



Cross-Section

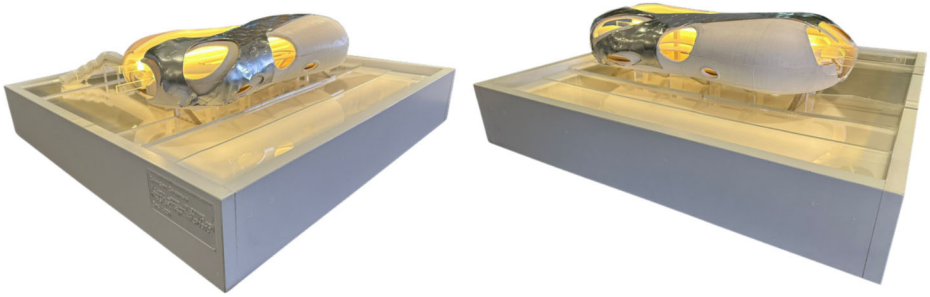


Level 1 - Lobby

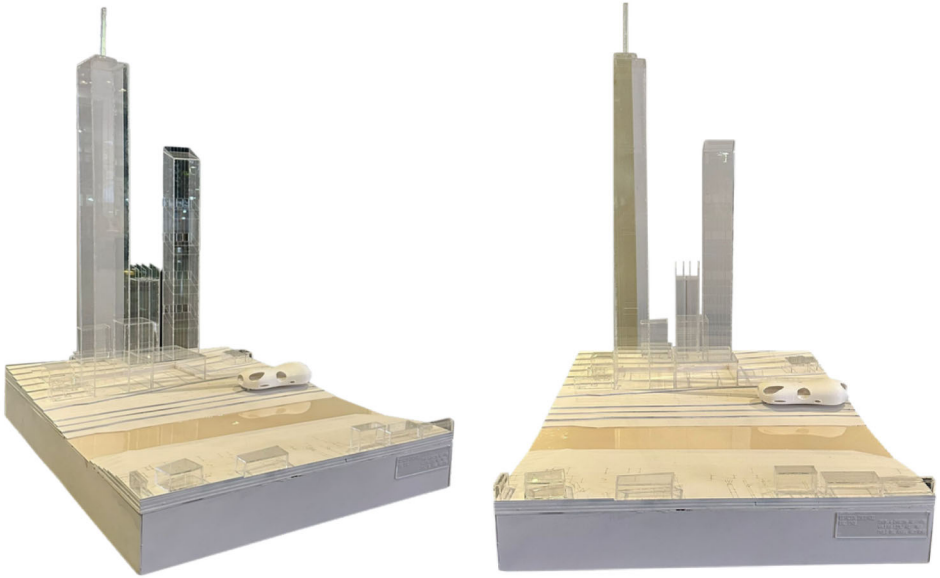


Level 2 - Platforms

Conceptual Model | Scale 1:500

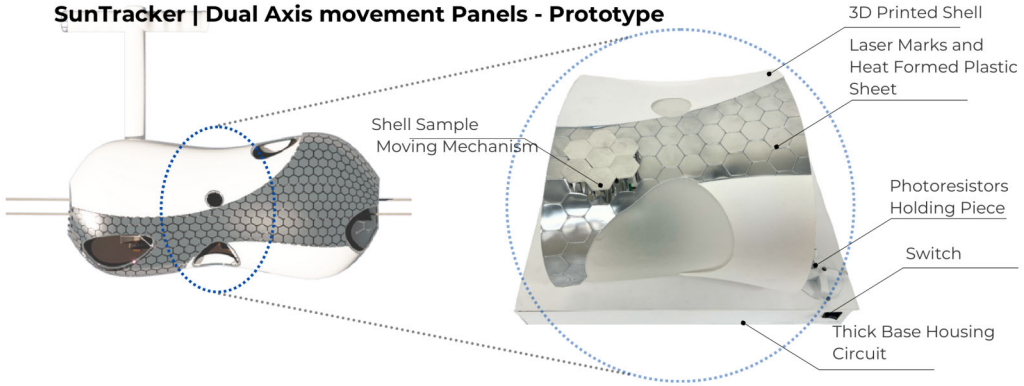


Final Model | Scale 1:250



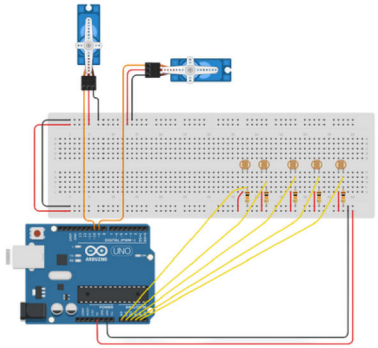
Model with Context | Scale 1:500

SunTracker | Dual Axis movement Panels - Prototype

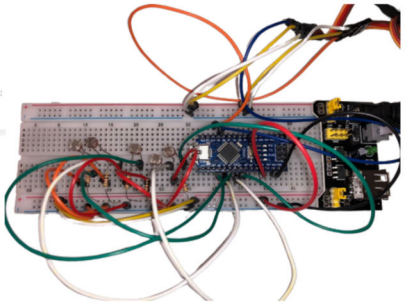


Chosen Section for Prototype Model

Model for Prototype



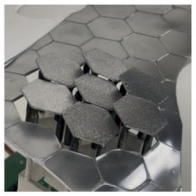
Arduino Code



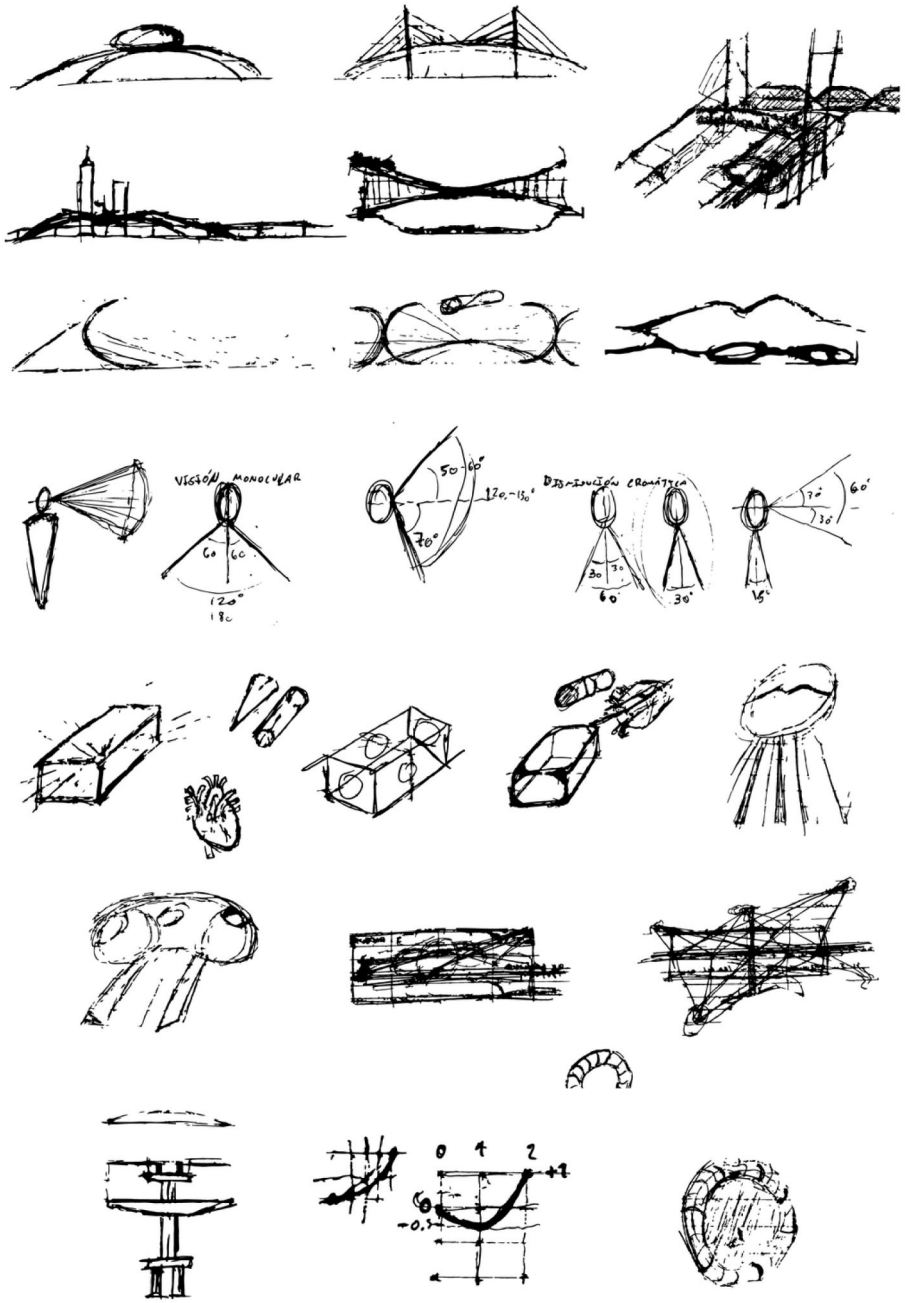
Arduino Circuit Simulation With Photoresistors to Track Light and Servos for Movement on Each Axis

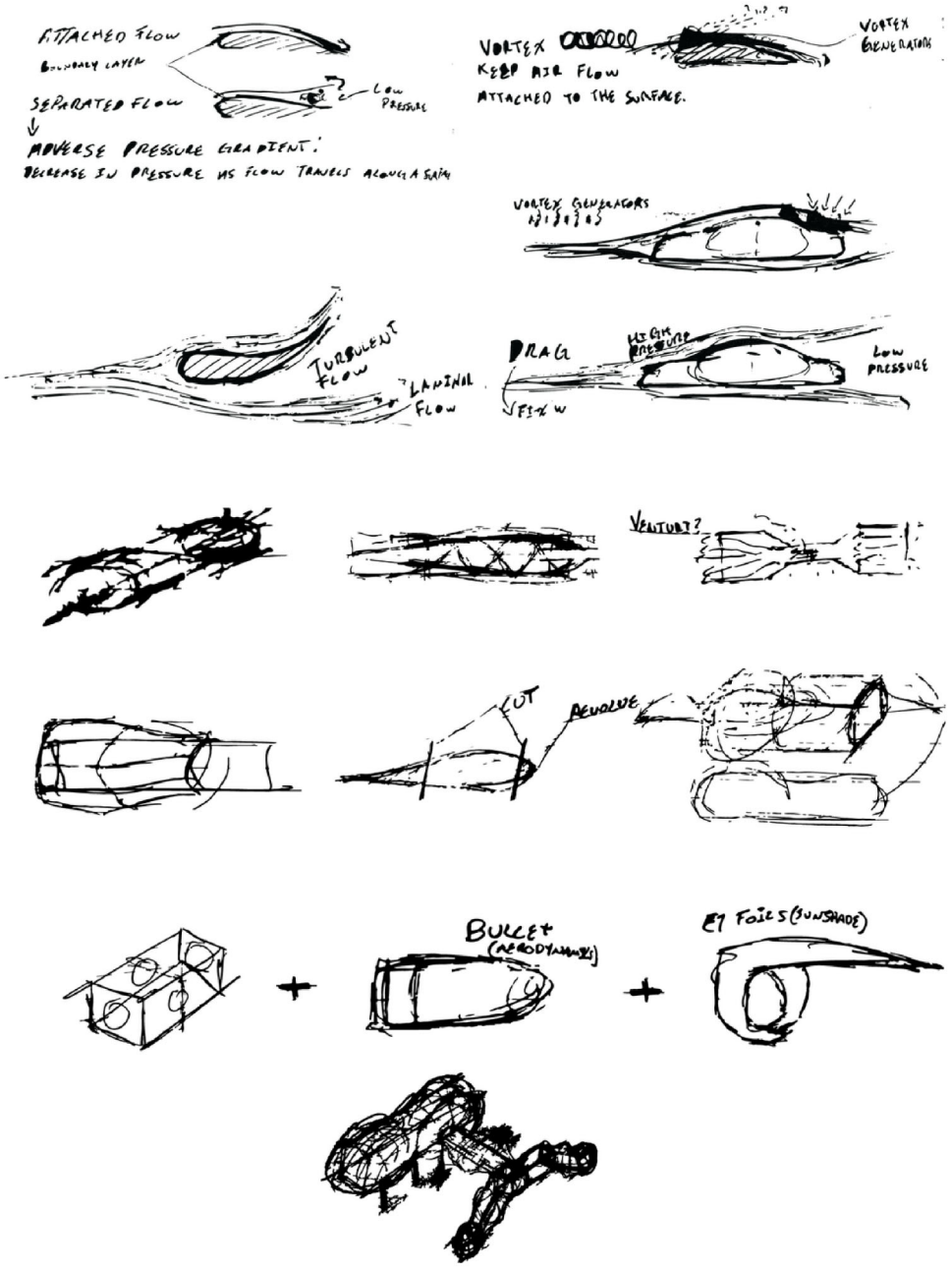
Arduino Circuit Test With Photoresistors to Track Light and Servos for Movement on Each Axis

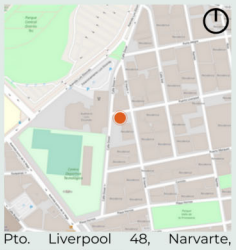
Final Working Prototype Model



Prototype Video Sample


VISIÓN MONOCULAR
120°
180
50-60°
120-150°
70°
DISMINUCIÓN CROMÁTICA
60°
30°
15°
0
1
2
+1
-0.5


ATTACHED FLOW
BOUNDARY LAYER
SEPARATED FLOW
LOW PRESSURE
ADVERSE PRESSURE GRADIENT.
DECREASE IN PRESSURE AS FLOW TRAVELS ALONG A SURFACE
VORTEX
KEEP AIR FLOW
ATTACHED TO THE SURFACE.
VORTEX GENERATORS
TURBULENT FLOW
LAMINAR FLOW
DRAG
HIGH PRESSURE
LOW PRESSURE
FLOW
VENTURI?
CUT
REVOLVE
BULGE+
(AERODYNAMIC)
+
+
E7 FOILS (SUNSHADE)



Pto. Liverpool 48, Narvarte, 64830 Monterrey, N.L.

流 Flow

"**Ryū**": The tower's ability to move, and change its components.

交 Exchange

"**Kō**": The tower's ability to swap and replace elements.

塔 Tower

"**Tō**": Representing the vertical structure itself.

Module Anchoring Animation

Colaborators:
Isabela Cárcoba
Mauricio Mandujano

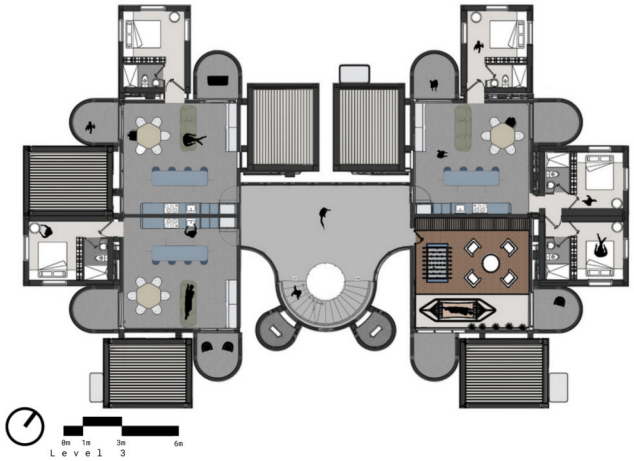
2023

Ryūkō Tō

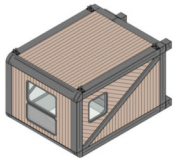
"The flow & exchange tower."

In the dynamic urban environment of Monterrey, N.L., the project emerges elegantly. Located in the vibrant Tec District, close to the prestigious University Tecnológico de Monterrey, this housing project stands as a masterful fusion between modernity and the influences of Japanese metabolic architecture. In this context, a contemporary oasis is born that not only embraces housing functionality, but also responds to the redensification of the area and enhances commercial growth.

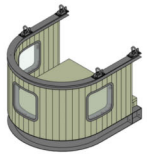
The modular tower, redefines the living experience through its constant movement and adaptability. Based on four main cores as a base. From level 1, the key spaces such as the kitchen, living room and dining room are replicated on each level of the cores, guaranteeing a uniform and efficient distribution.



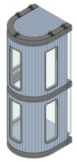
The true versatility of RYUKO lies in the possibility of exchanging room modules, balconies, stairs (for duplexes) and bathrooms (for "rooftops"). To facilitate this flexibility, a fixed crane on the main distribution tower allows, with the help of an operator, the placement and exchange of modules without the need for external machinery, ensuring an agile and uncomplicated process.



Bedroom Module

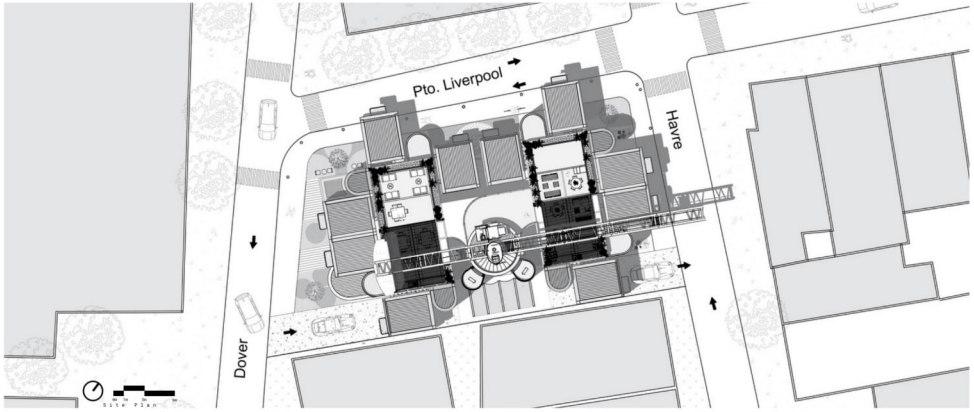


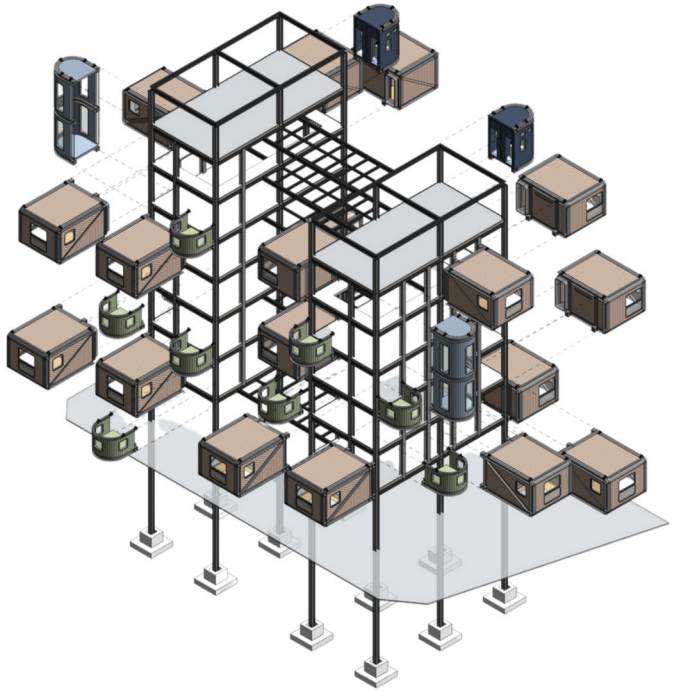
Balcony Module



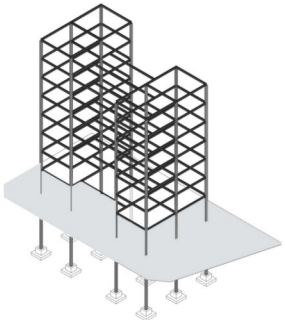
Duplex Staircase Module

Rooftop Restrooms



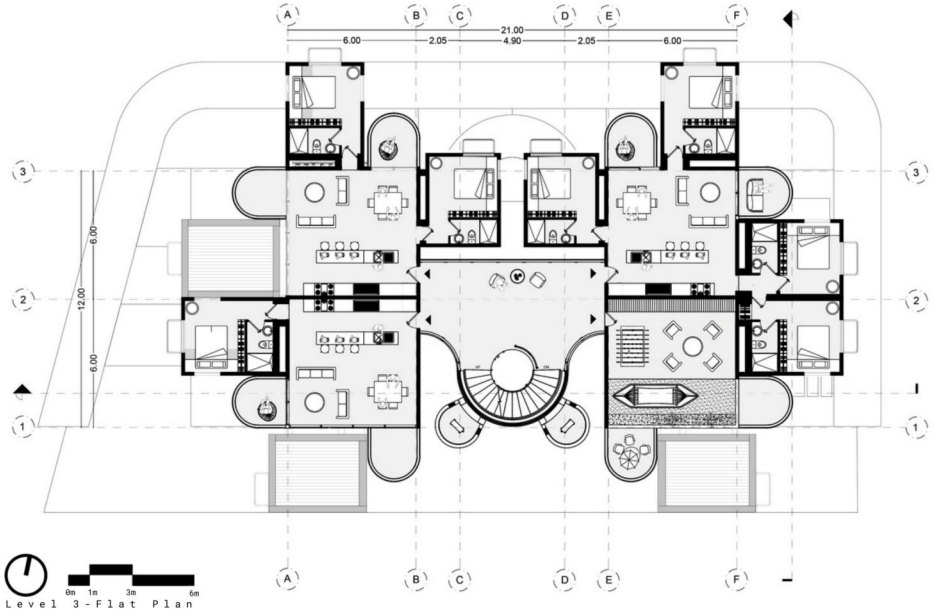


Exploded Isometric - Module Ensemble

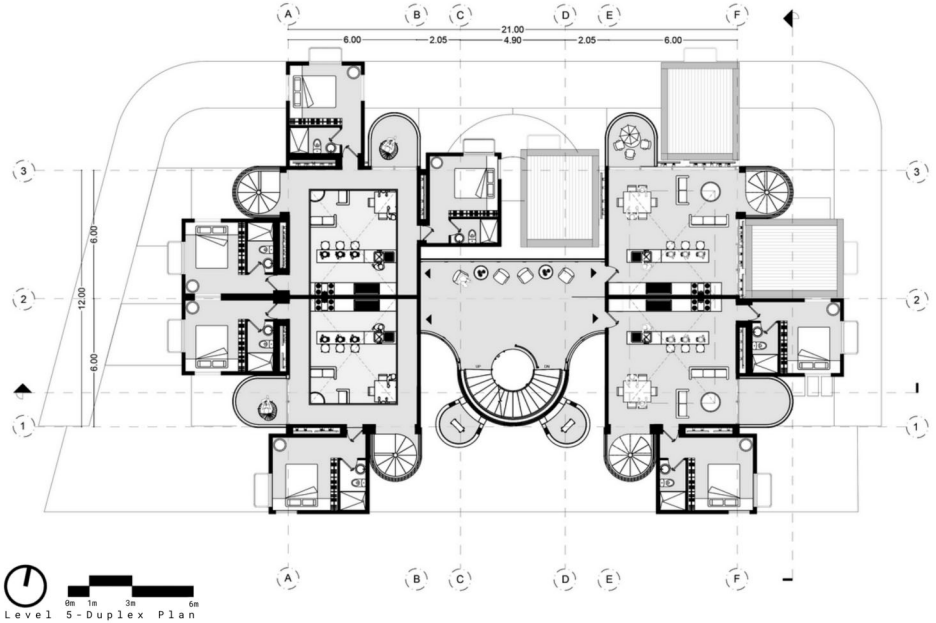


Strucural Isometric

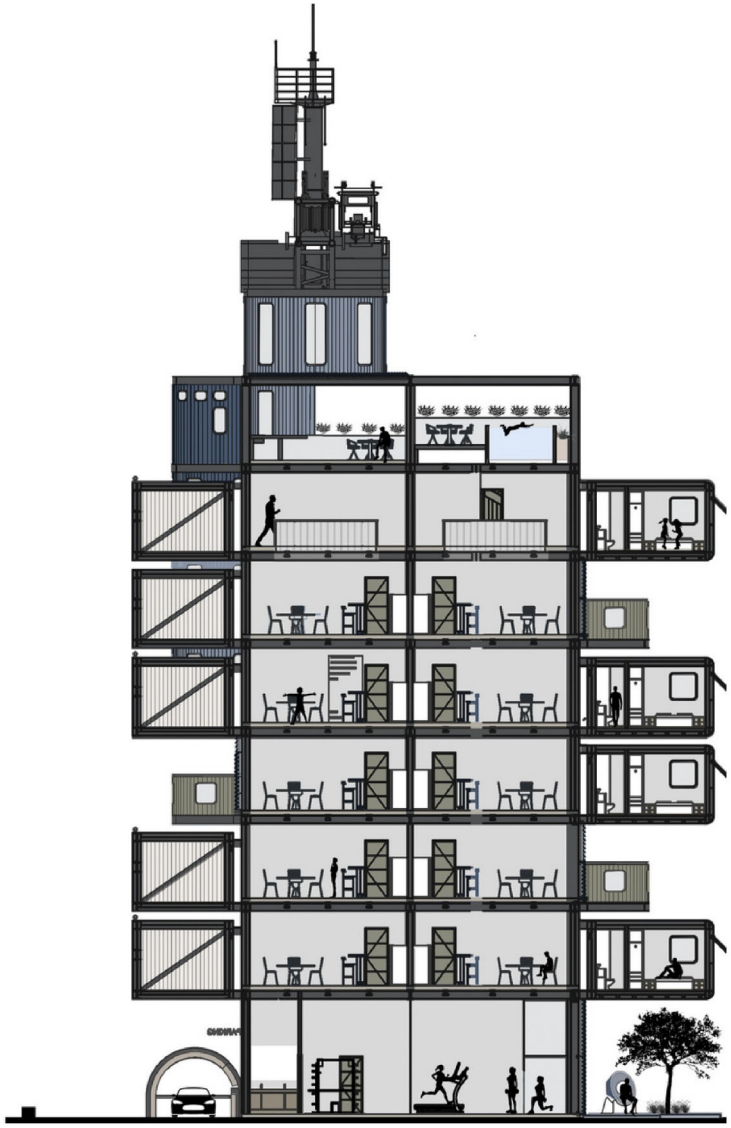
The conceptual core revolves around efficiency through modularity and prefabrication. The inspiration is found in the streamlining of the construction process and the versatility of the design. Prefabricated elements, together with the strategic use of steel, allow rapid deployment and constant readjustment of modules even in diverse environments.



Level 3-Flat Plan

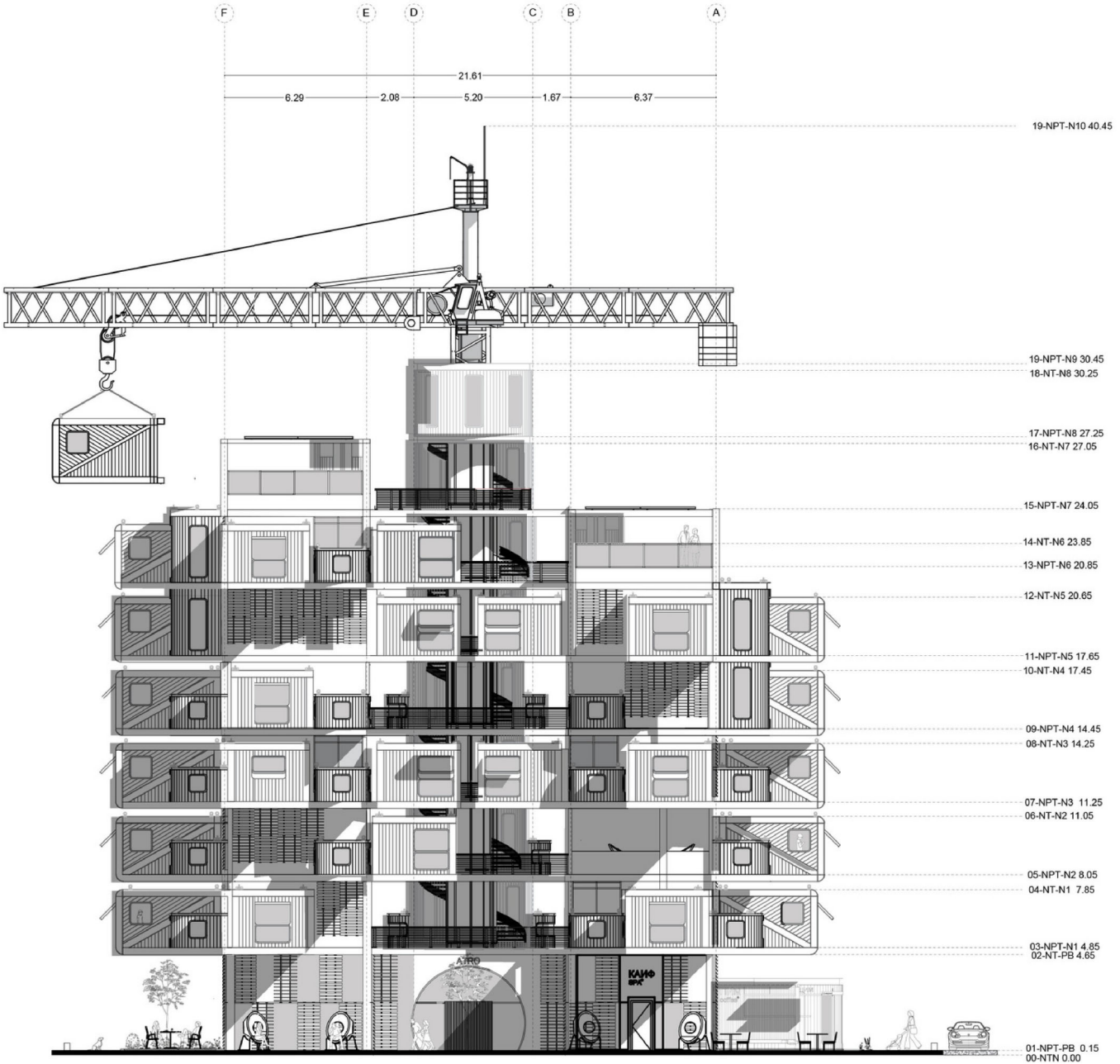


Level 5-Duplex Plan



0m 1m 3m 6m

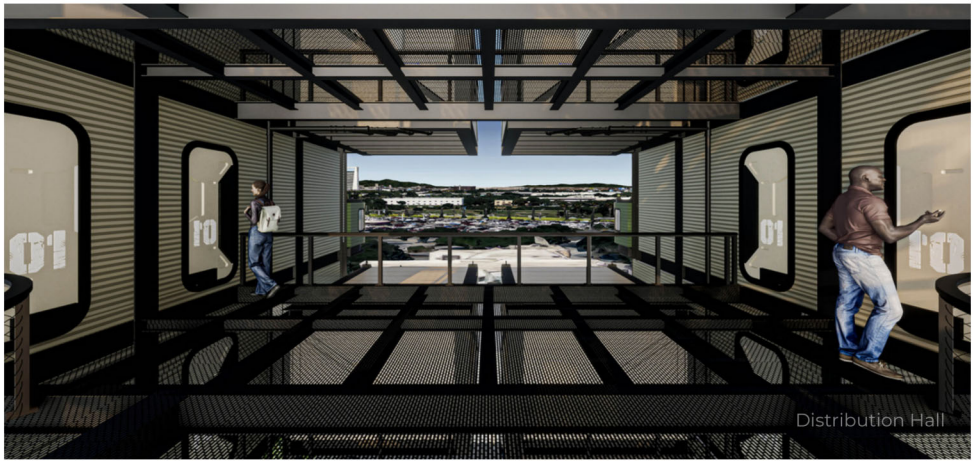
Cross Section



0m 1m 3m 6m

North Facade

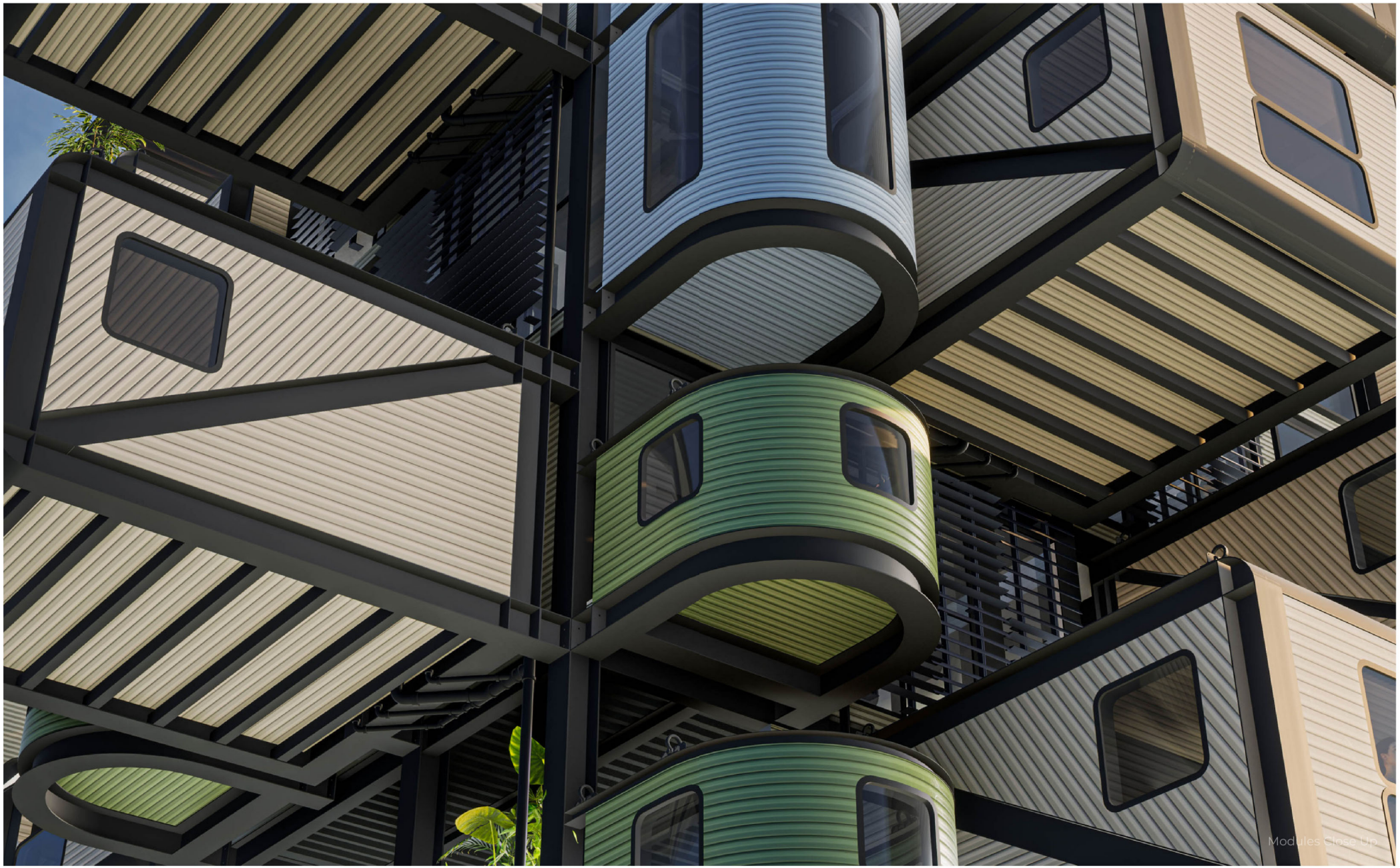
Rooftop

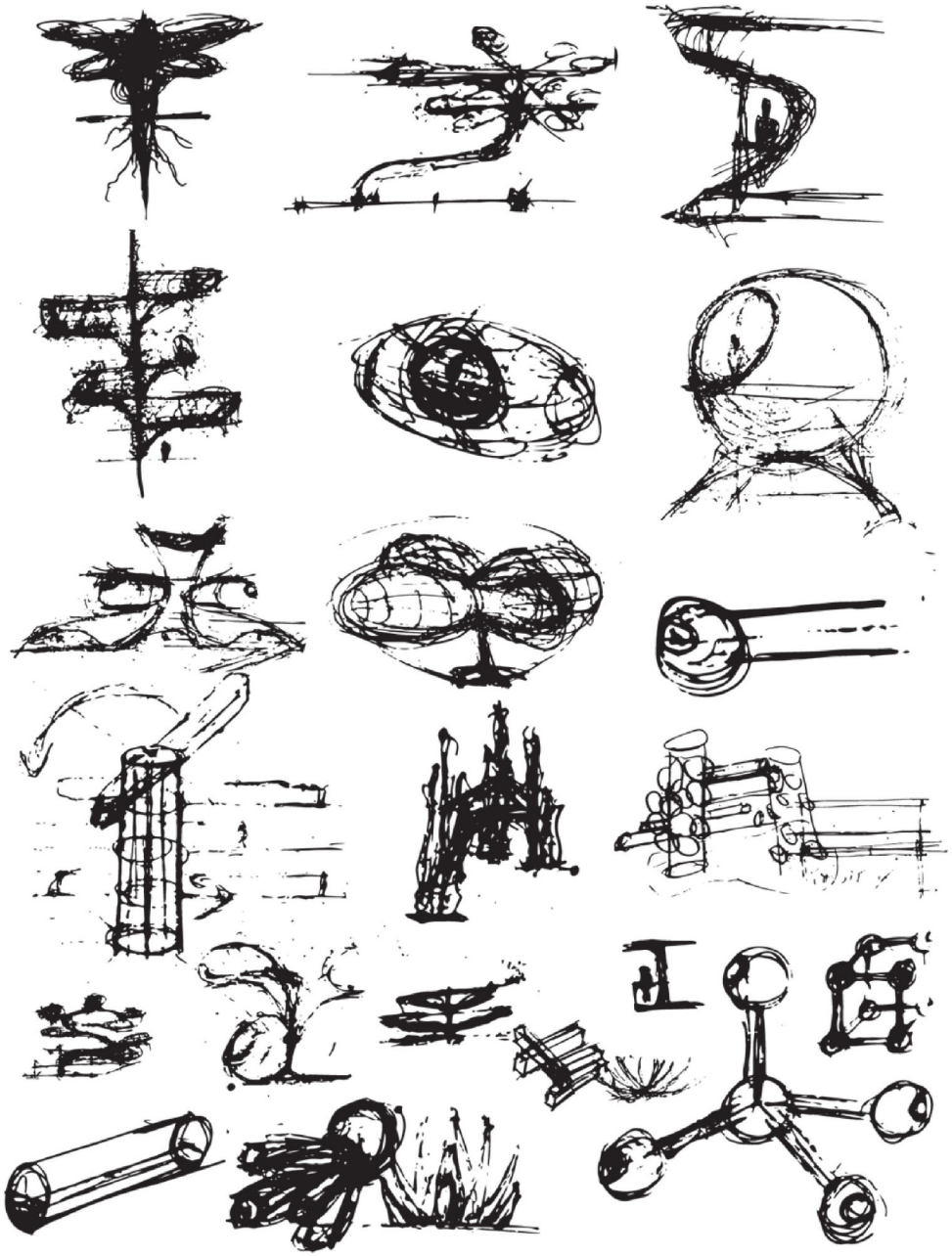

Distribution Hall

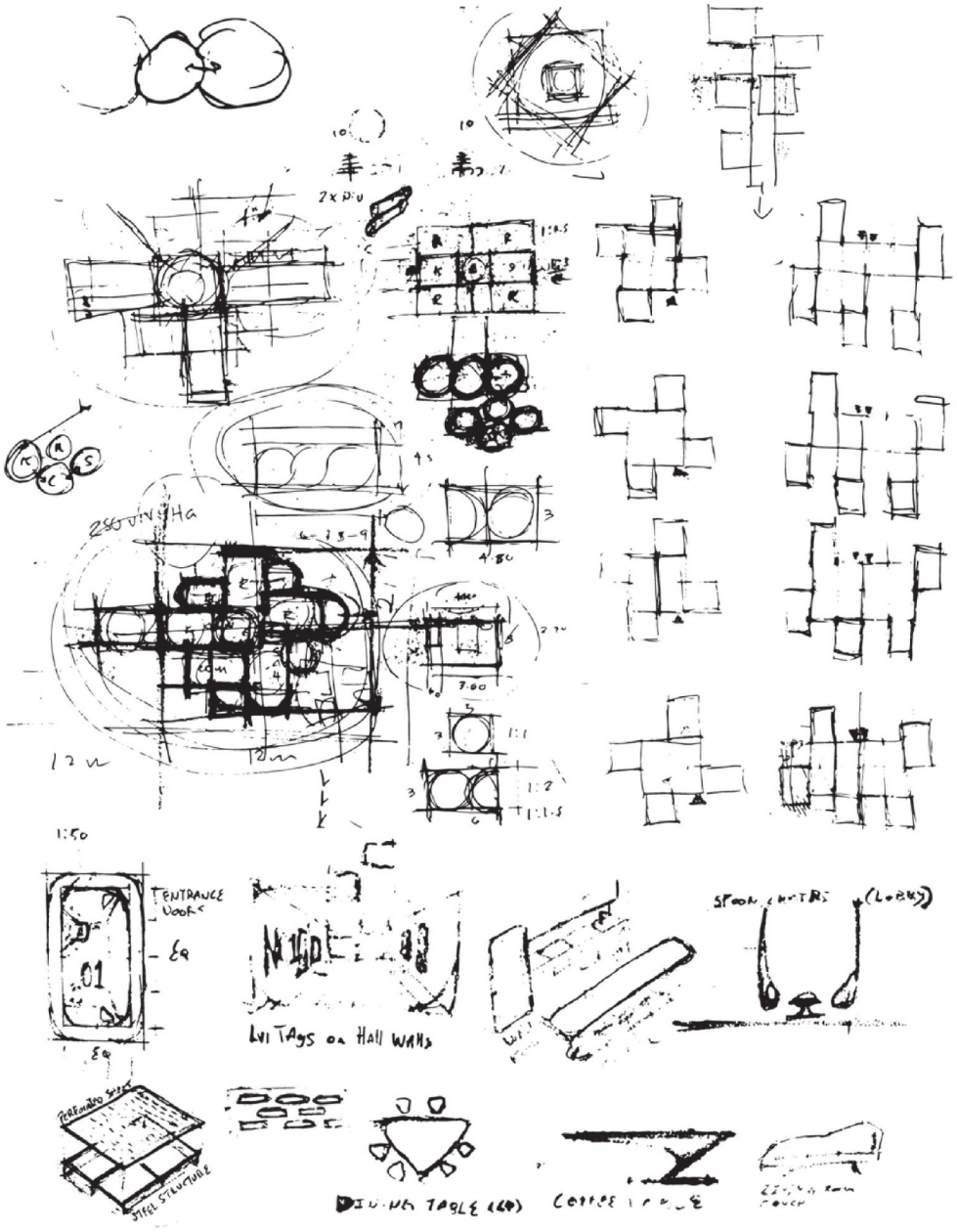
RYUKOTO
КАИФ
SPA
ПИМ
coffee
Commercial Area

Duplex Interior

Flat Interior


Modules Close Up

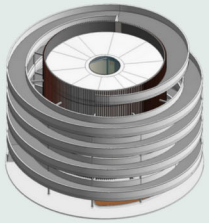



1:50
ENTRANCE DOOR
01
LVI TAGS on Hall WALLS
SPOON (WATER) (LOBBY)
DINING TABLE (60)
COFFEE TABLE

Sin Nombre de Col 35, Berel LT26, Ladrillera, Monterrey, N.L.

Helix Vitae

"Helix Vitae", a Latin expression meaning "Spiral of Life", is an evocative name for the project, as it symbolizes an uninterrupted cycle of life with a beginning and an end. Just like in an upward spiral, each phase of the project is a discovery and opportunity for growth. Reaching the top reveals a stunning view, just as in the final stages of our building where, despite less activity, a unique depth and richness is experienced. "Helix Vitae" represents not only the constant evolution of life, but also the narrative of our architectural project.



Colaborators:
Isabela Cárcoba

2023

Helix Vitae

"Spiral of life."

This project is located in Monterrey, Nuevo León, and is located in an unusual place: a pantheon. What makes it particularly interesting is the client's initiative, which seeks to revitalize and connect this land with its surroundings and with the city as a whole.

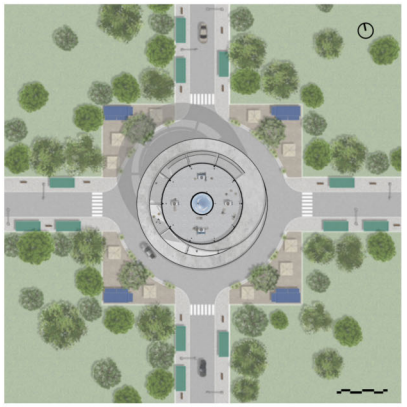
To achieve this, a unique plan has been conceived which involves the creation of a promenade incorporating commercial areas and leading directly into the heart of the land. At the center of this site stands an iconic building designed to attract local residents and those in the vicinity. The idea is to transform a space that is traditionally associated with the commemoration of the deceased into an active meeting point, where daily life and commerce intertwine, encouraging the community's reconnection with this place in a unique and meaningful way.

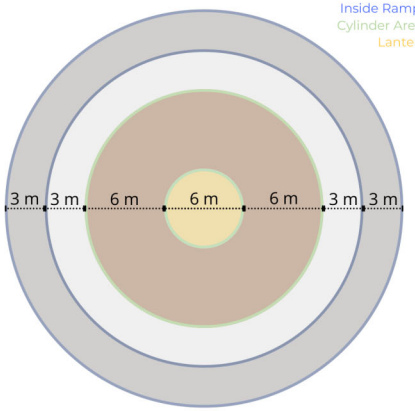
The project's design takes into account the site's history and cultural significance, seeking to respect and enhance its existing features. The iconic building's design incorporates modern and innovative elements. The result is a harmonious blend of the old existing context and the new, creating a space that is both functional and aesthetically pleasing.

The promenade will be lined with trees, providing shade and a welcoming atmosphere for visitors. The commercial areas will feature a variety of shops and restaurants, catering to different tastes and preferences. The aim is to create a vibrant and lively space that will attract people from all walks of life and become a hub for community activity and interaction.

Overall, this project represents a unique and innovative approach to urban revitalization, transforming a traditional space into a modern and dynamic one. It is a testament to the power of creativity and imagination in shaping our built environment and our communities, and a reminder of the importance of preserving and celebrating our cultural heritage.

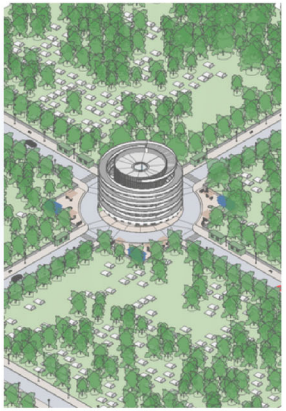
Site Plan

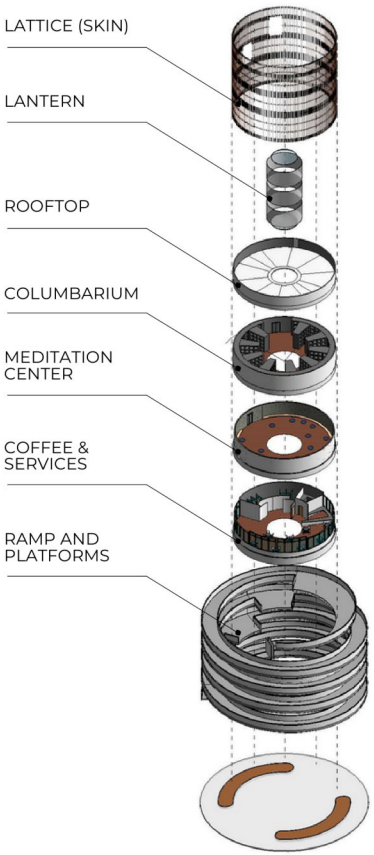


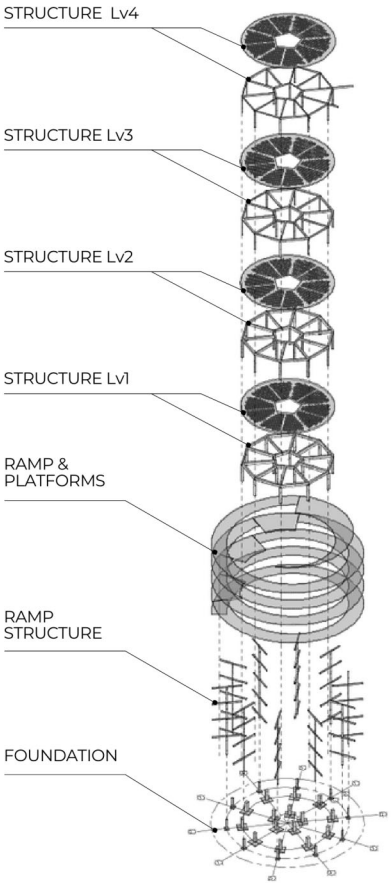


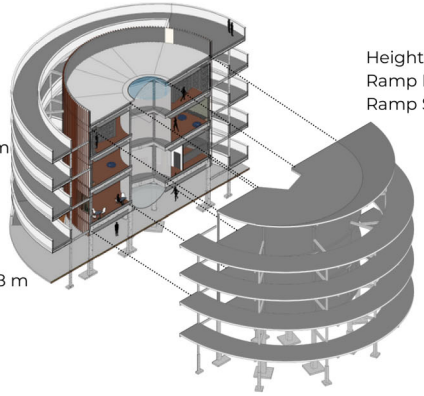
General Dimentions

Cross Section







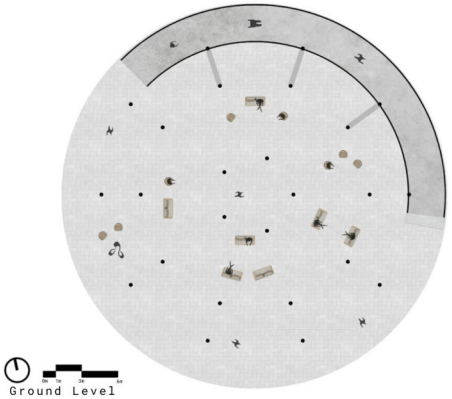


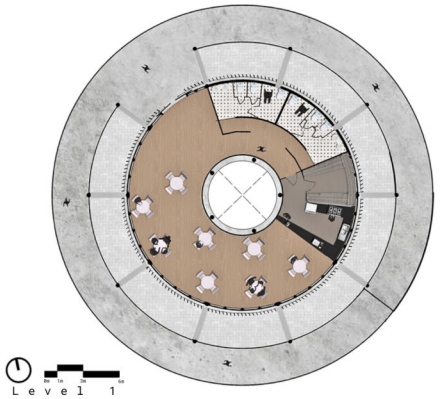
Height Lv2 to Lv3 = 4.2 m
Ramp Length Lv2 to Lv3 ≈ 84.8 m
Ramp Slope ≈ 4.9%

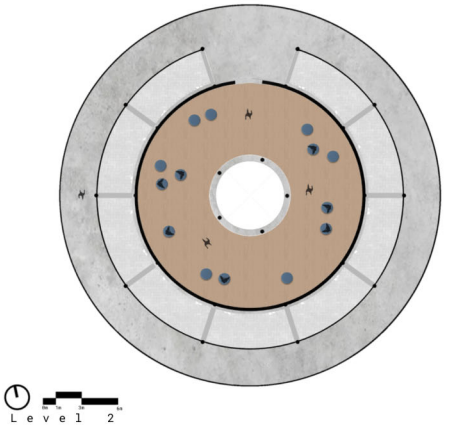
Height GndLv to Lv1 = 4.7 m
Ramp Length GndLv to Lv1≈ 84.8 m
Ramp slope ≈ 5.5%

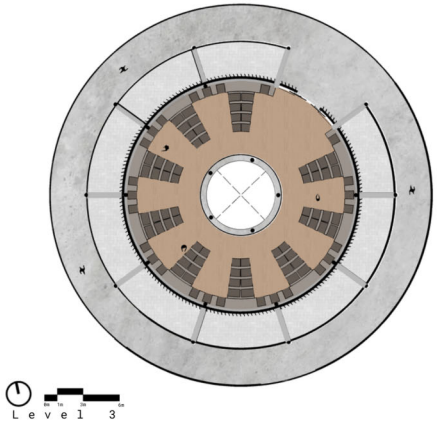
Height Lv3 to RoofTop= 4.2 m
Ramp Length Lv3 to RoofTop ≈ 84.8 m
Ramp Slope ≈ 4.9%

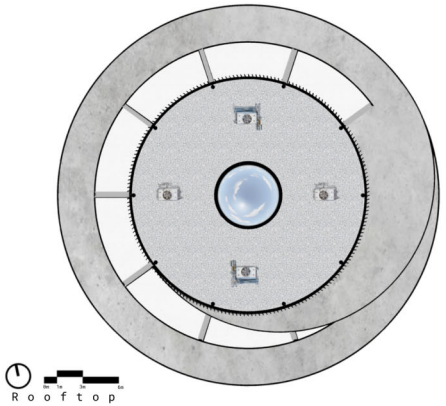
Height Lv1 to Lv2 = 4.7 m
Ramp Length Lv1 to Lv2 ≈ 84.8 m Ramp Slope ≈ 5.5%



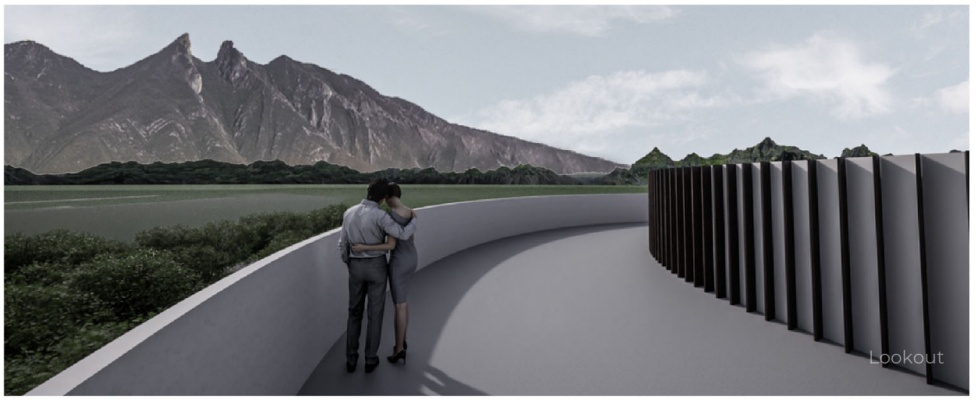








Columbarium


Lookout

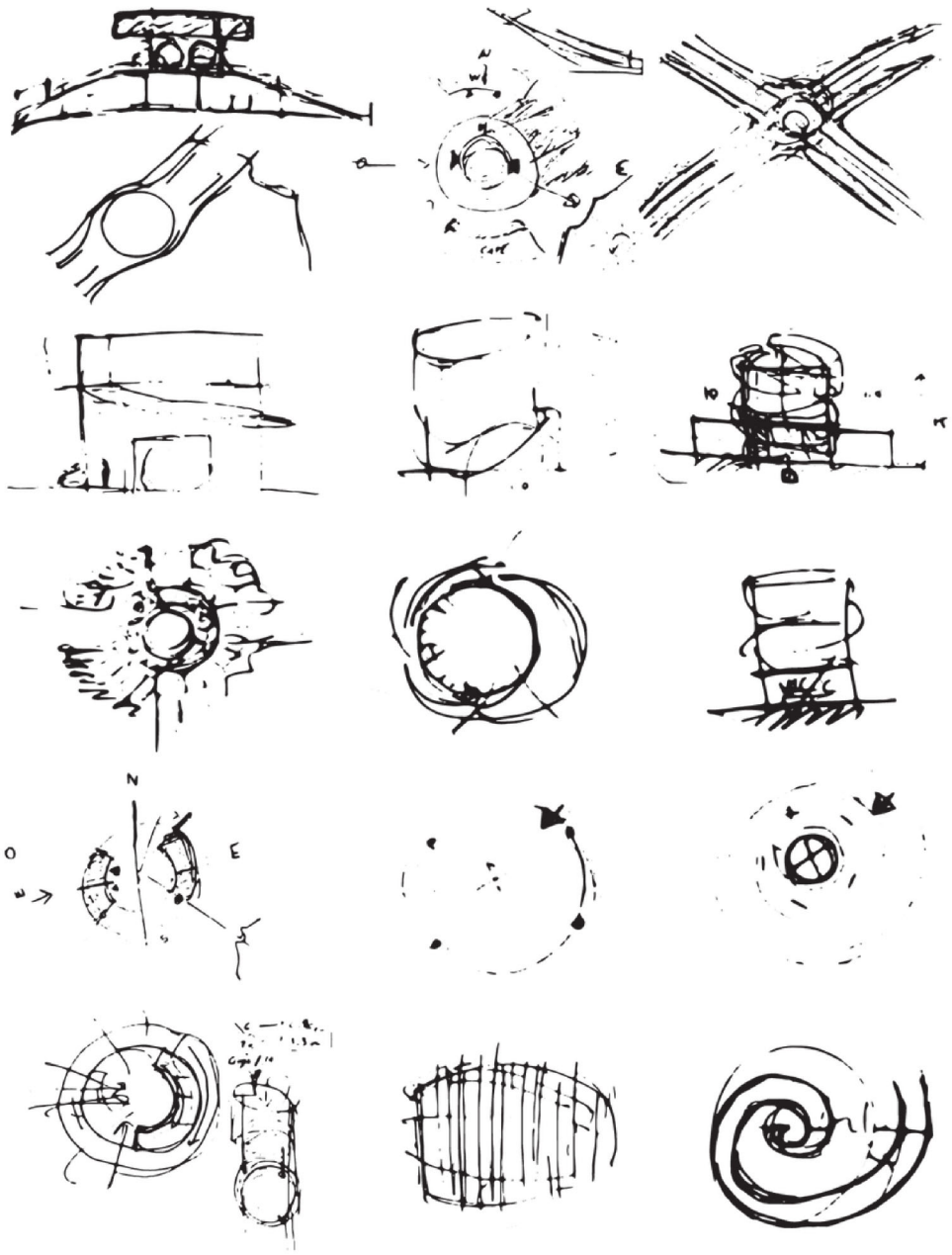
Meditation Center

Ramps

Coffee & Services

Ground Level

Main Facade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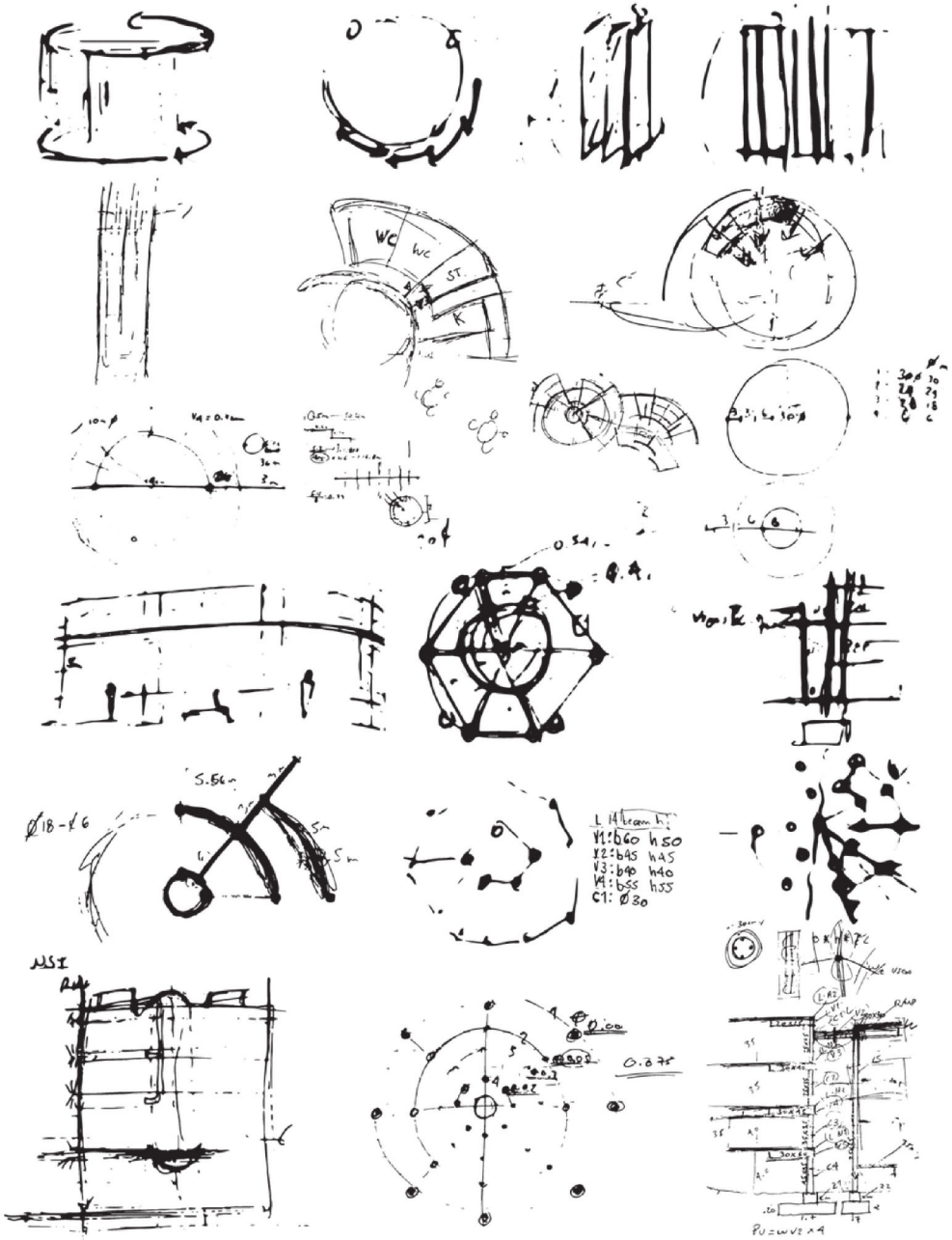
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WC
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S.56m
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L 14 beam ty
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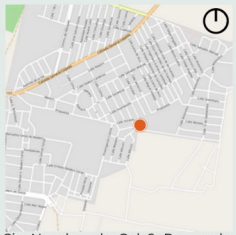
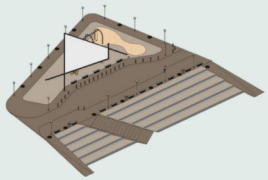
V2: b45 h45
V3: b40 h40
V4: b55 h55
C1: Ø30
NSI
0.875



Sin Nombre de Col 6, Parras de la Fuente, Coah.

DINO PLAZA

The name "Dino Plaza" for the project finds its justification in the rich paleontological history of Parras de la Fuente, where ancient dinosaurs left their mark in the prehistoric past. This name not only pays homage to those ancient inhabitants, but also evokes curiosity and imagination by creating a bridge between the past and the future. In addition to being a symbol of the connection between history and modern development of the city, "Dino Plaza" becomes a space that encourages exploration and learning, inspiring residents and visitors to reflect on the enriching geological history of the city, region and its relevance in the current identity of Parras de la Fuente.



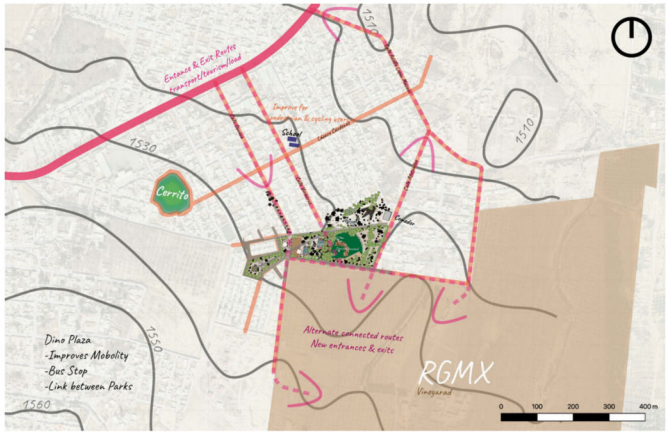
Colaborators:
Isabela Cárcoba
Mauricio Mandujano

2022

Dino Plaza

"Traces of the Past, Path to the Future"

The Dino Plaza project, a vital part of the ambitious Project Paws plan, stands in the beautiful enclave of Parras de la Fuente, a town rich in history and rooted in wine culture. This proposal is part of a broader effort, aimed at charting a path of comprehensive development in the Angeles area, addressing the realities and challenges of the community to shape an intervention that is genuinely participatory and transformative. In this context, Dino Plaza emerges as a cornerstone destined to raise the quality of life of the inhabitants and enhance the tourist experience in the city, focusing on the urgent need to have first-class recreational, sports and social spaces in this region. of great winemaking significance.



"...we do not have transportation to take us and bring us from the area to the center..."

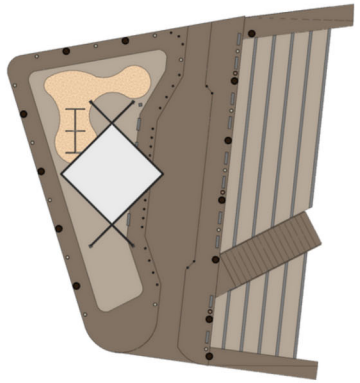
"...as there are many children here, we need to have a playground..."

"...we all want to be part of the magical town, we want to belong..."

(From interviews while visiting the site.)

The essence of Dino Plaza lies in its ability to perfectly merge the historical legacy of Parras de la Fuente with the architectural avant-garde that it seeks to promote. Strategically located at the top of the park, where Oaxaca Street culminates in the park's horizon, this stop not only reinforces the school transportation system, but also becomes a landmark, a beacon that welcomes visitors and locals equally.

Dino Plaza is not simply a physical place, but a window into education and imagination. While children wait for school transportation, this plaza becomes a space for exploration, a place where curiosity is encouraged and young minds are nurtured. Unconventional shapes and contrasting materials invoke innovation and progress, inspiring future generations to transcend the known. Additionally, Dino Plaza could even host local commerce opportunities, a place where artisans and merchants can interact with residents and tourists alike, creating a vibrant hub of economic activity.



Dino Plaza, within the Project Paws in Parras de la Fuente, transcends mere architecture to become a testimony of the vision and commitment to transform a community. Through innovation and respect for local history, the square becomes a multigenerational meeting point, a catalyst for unforgettable experiences and an icon that celebrates both the present and the future of Parras de la Fuente. From its prominent location at the top of the park, Dino Plaza offers a panoramic view of the city and its vineyards, embodying the dynamic and diverse spirit of this corner of Mexico, where architecture and wine culture converge in a unique expression of identity and growth.

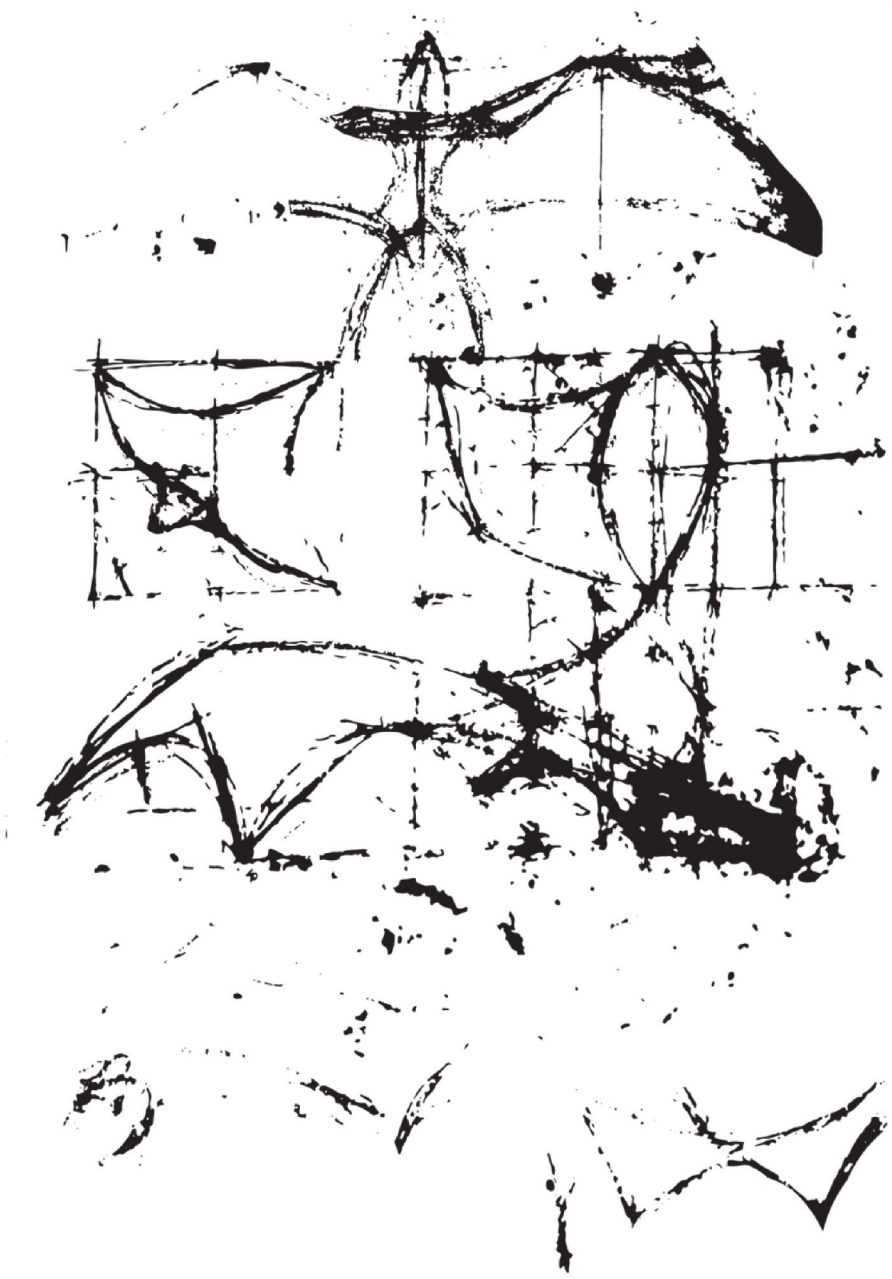
Aereal View

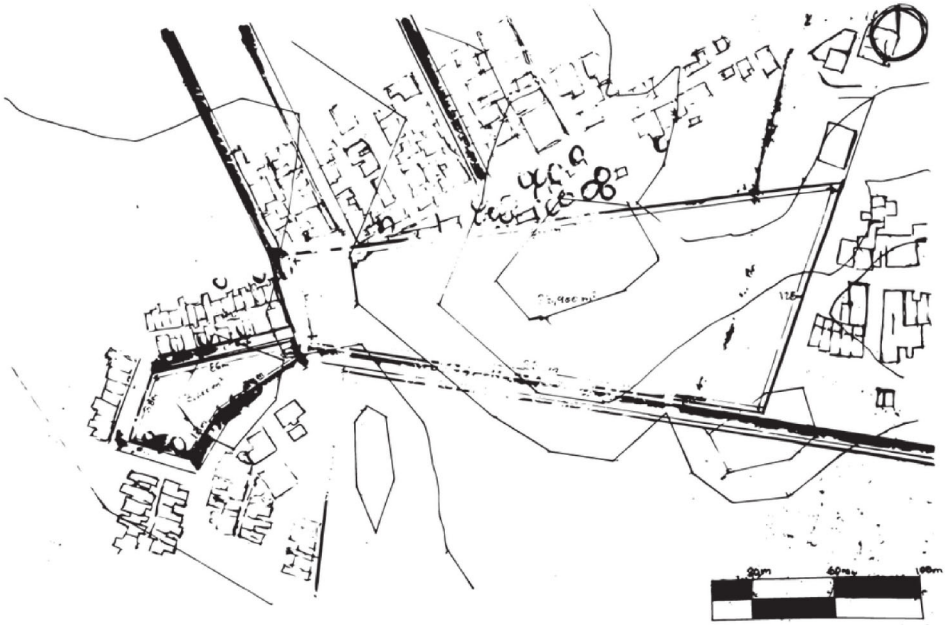
Touristic Spot

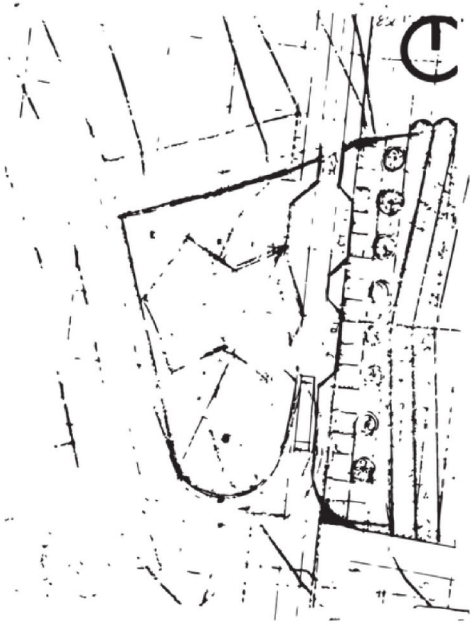
Line 74
Bus Stop

Playground

Top View



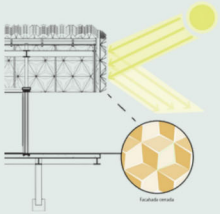


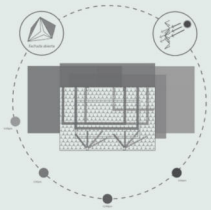


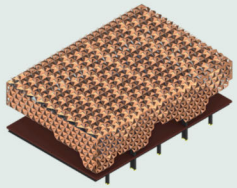
Rufino Tamayo Park, Av. Rufino Tamayo 100, Corporativo Prodesa, 66278 San Pedro Garza García, N.L.

RT RUFINO TAMAYO

PAVILION







Colaborators:
Isabela Cárcoba
Valeria López

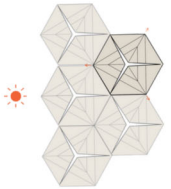
2024

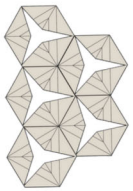
RT Pavilion

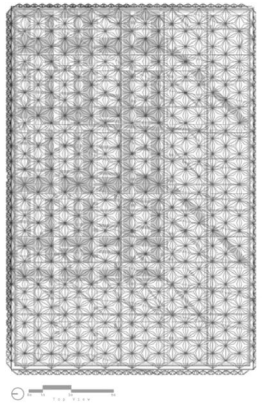
"Redefining the Traditional Pavilion"

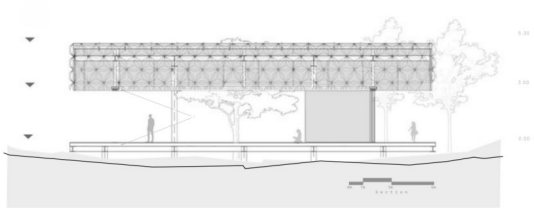
OTIIMA 2024 Mexico Pavilion, is an architectural challenge that merges the innovation of OTIIMA window systems with the creativity of architecture students from Nuevo León. This project arises from the collaboration between OTIIMA Systems, the Colegio de Arquitectos de Nuevo León, AC, and San Pedro Parks, with the aim of promoting excellence in architectural design and spatial expression. In response to the contest's call, we have conceived a 300m2 pavilion that goes beyond being simply a physical structure; it is a dynamic and multifunctional space that invites the community to explore the concept of transparency as a fundamental element of contemporary architecture.

The photosensitive panels open and close in response to sunlight. When sunlight is abundant, the panels open to let natural light in, creating a welcoming atmosphere. Conversely, they close partially or completely in intense sunlight or low light conditions to regulate light levels and enhance comfort. This adaptive feature not only improves energy efficiency but also fosters a seamless connection between indoor and outdoor spaces, enhancing the overall experience of the pavilion.

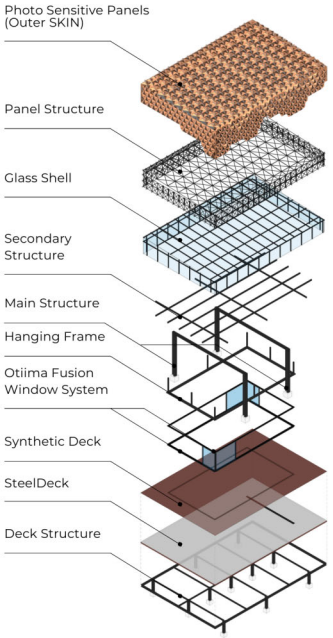


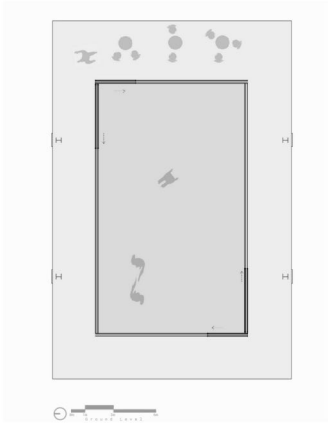




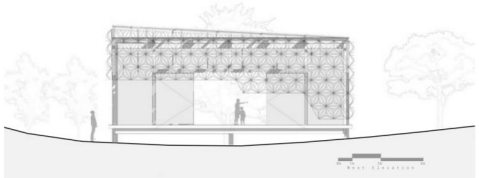


The pavilion's primary framework consists of four robust steel columns and two beams. These components create two distinct goalframe-like structures, each supporting a network of smaller beams that extend in perpendicular planes. This secondary framework serves a crucial role in providing support for the outer panels and glass shell of the pavilion. Suspended from this secondary structure is a steel frame, strategically positioned to enable an expansive, nearly fully open level within the pavilion. This frame serves as a base for mounting the upper railings necessary for the sliding doors, ensuring both structural integrity and functional design.





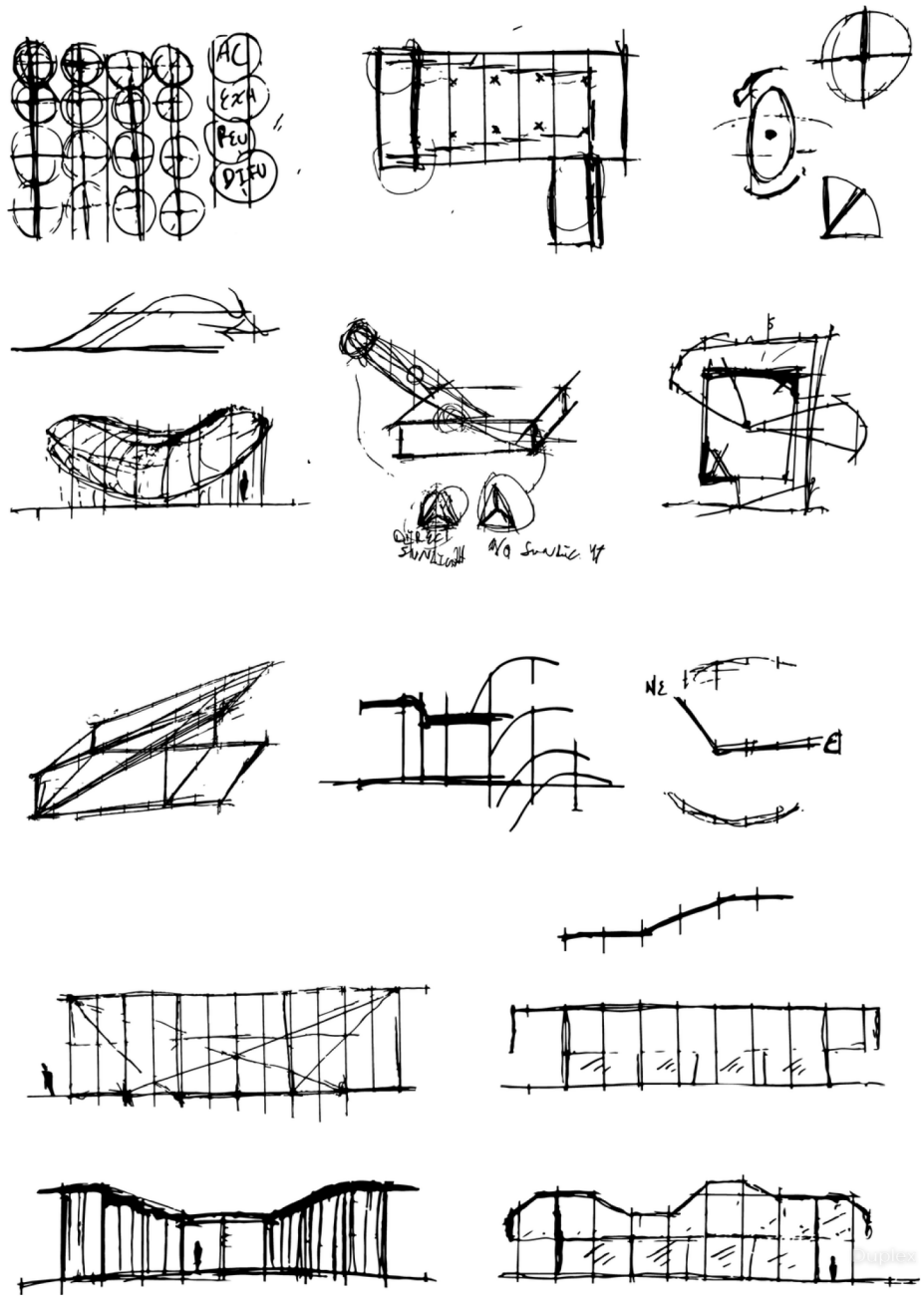
The FUSION 1.0 window system by OTIIMA integrates each of the project's components and enables the implementation of solar-adaptive systems as part of the architectural experience. This complex system, incorporating thin frames reinforced with polyamides and resistant glass, allows the pavilion to be transparent and permeable to its natural surroundings without sacrificing its thermal condition when closed.



Pavilion Interior

East Side View

North-West View



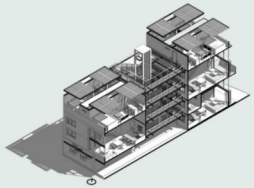
Independencia 221, Casco Urbano, 66200 San Pedro Garza García, N.L.

Doppio

(Italian for double.)
/dóp·pio/

1. Che è due volte la dimensione o quantità normale o consueta.

The design focuses on the concept of "double height", a defining feature that adds spaciousness and elegance to each living unit. More than just a structure, Doppio aspires to be a space that fosters interaction and community, while providing each resident with their own private envirnoment.



Colaborators:
Isabela Cárcoba
Mauricio Mandujano

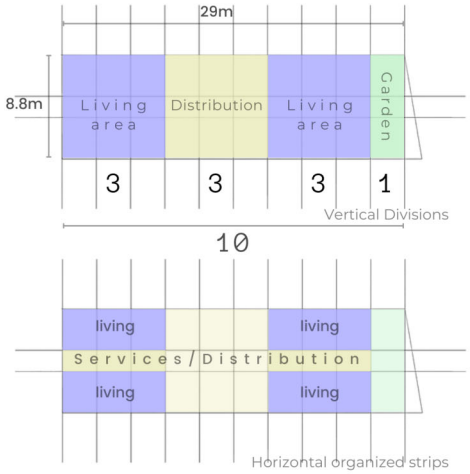
2023

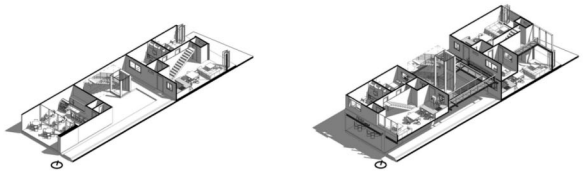
Doppio

"Double Height, Double Life."

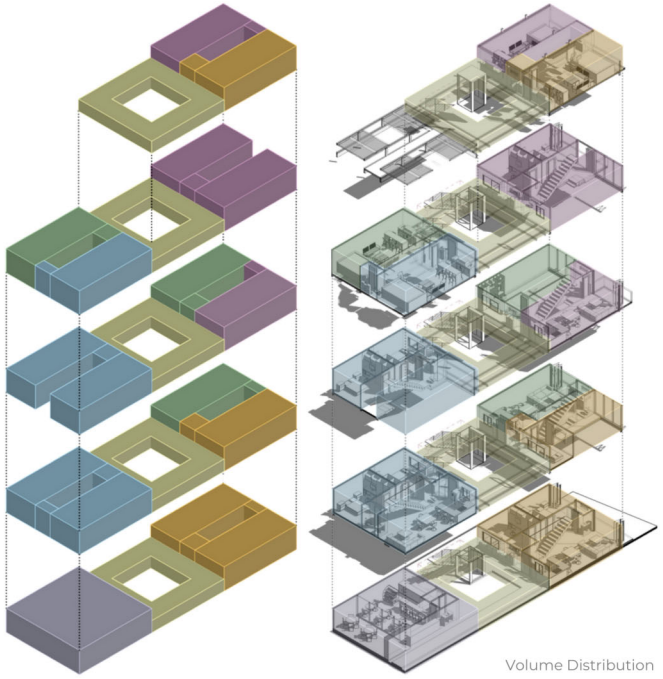
The Doppio Project emerges as an innovative collective housing proposal located in the Urban Center of San Pedro Garza Garcia, N.L. Sandwiched between two existing buildings, this project transforms a narrow but elongated plot of land of approximately 255m2 into an oasis for living and functionality. Doppio not only reimagines residential architecture, but also adds a commercial component on the ground floor, fusing modern urban living with comfort and contemporary aesthetics.

The land is divided equally into ten vertical strips, giving shape to three main areas. Two of these areas, located at the ends, are intended for individual living areas, while the central strip acts as a shared distribution space for both areas. In addition, a green area is reserved as a private garden for the home on the ground floor, infusing the essence of nature into the urban environment.





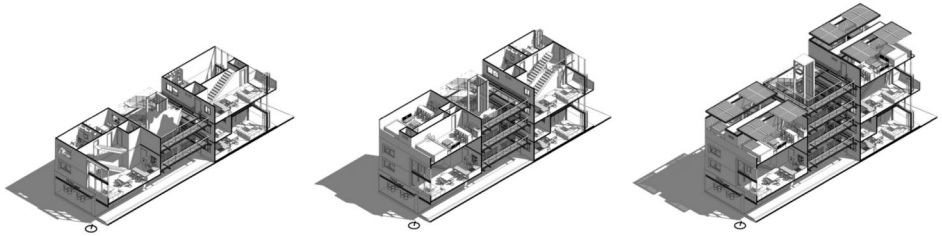
The horizontal organization of the project follows a similarly balanced strategy. They are divided into three strips, two of which are exclusive living spaces for the homes, where the double height creates a feeling of expansion and luminosity. The central bay, on the other hand, combines circulation and service areas, guaranteeing functional efficiency without compromising aesthetics.



Volume Distribution

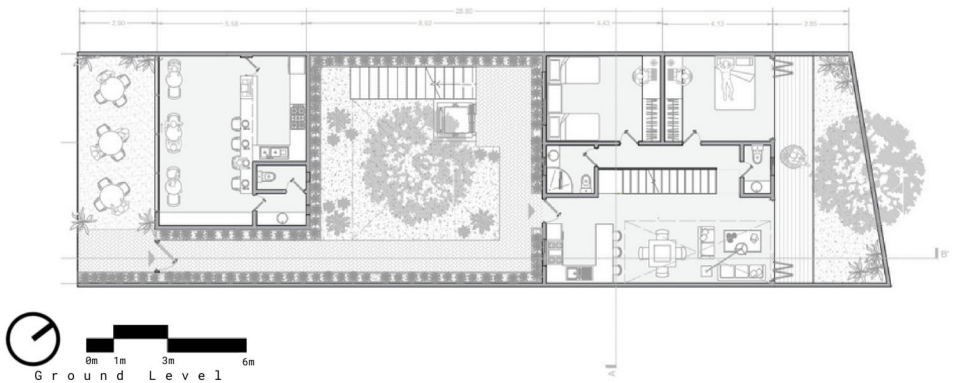
In addition to the housing units, Doppio dedicates the last level of each division to individual social areas per home. These spaces encourage social interaction in a vertical environment. Here, residents can enjoy panoramic views of the city and experience a sense of place beyond their private units.

The Doppio Project is not just an architectural design, but a reinterpretation of modern urban life. Through the intelligent use of space, double height, flexibility and attention to details, this project seeks to improve the quality of life of its inhabitants and enrich the architectural identity of San Pedro Garza Garcia, N.L. Doppio fuses functionality, aesthetics and community in an architectural work that transcends the conventional limits of collective housing.

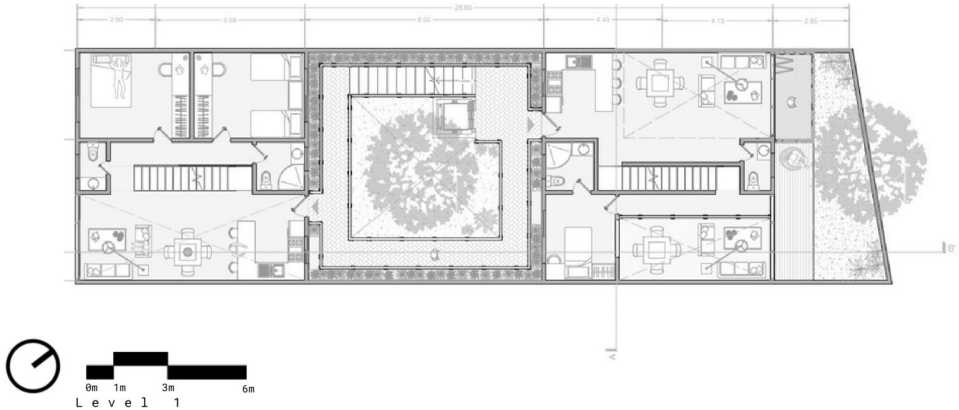


Isometric Views by Level

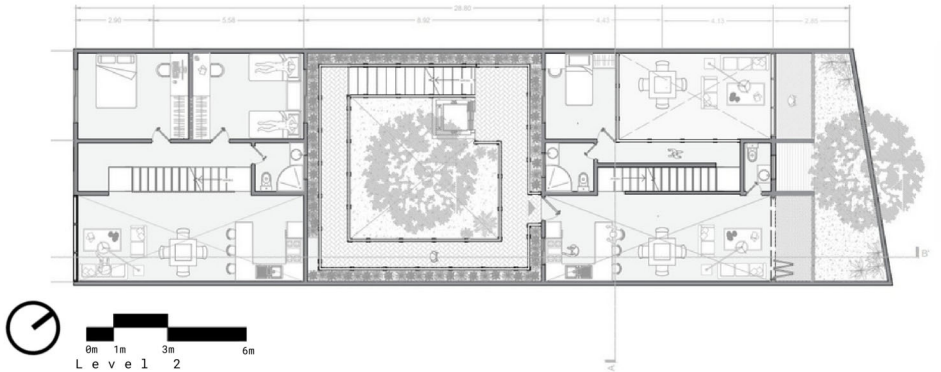
Doppio is distinguished by its focus on flexibility and diversity. Each home not only shares a cohesive aesthetic, but is also tailored to individual needs. The subsequent homes displace volumes to generate variations of housing units with different accommodation capacities. These variations maintain the double height as a unifying element, guaranteeing a unique residential experience in each case.



Ground Level



Level 1



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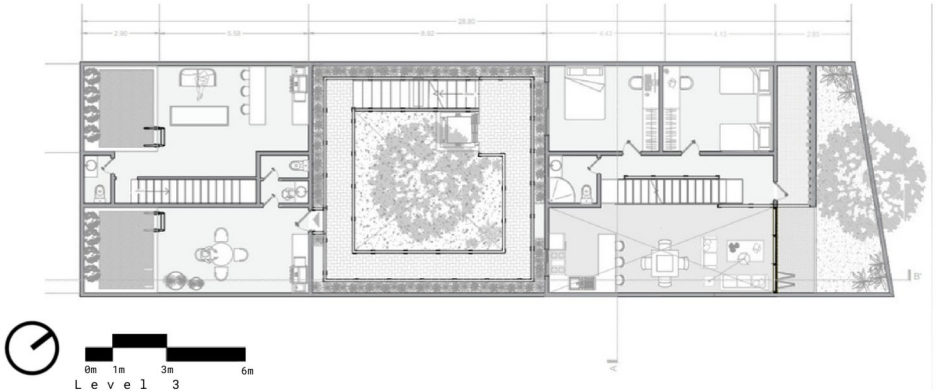

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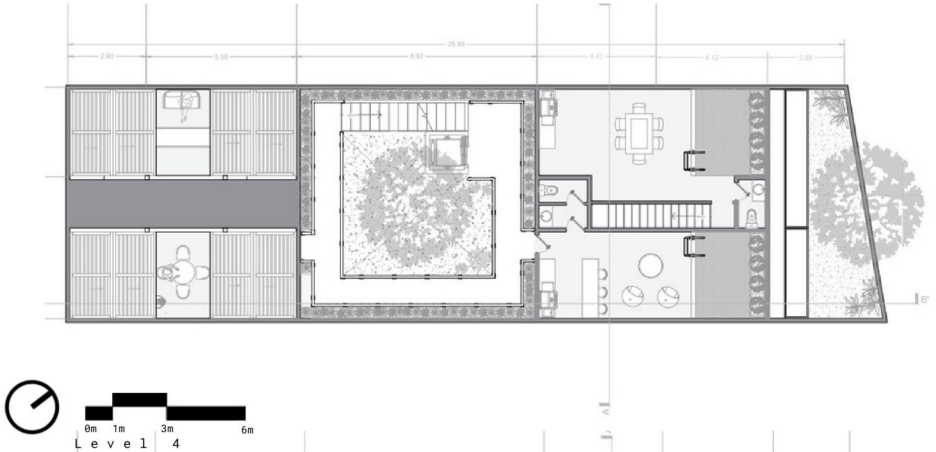

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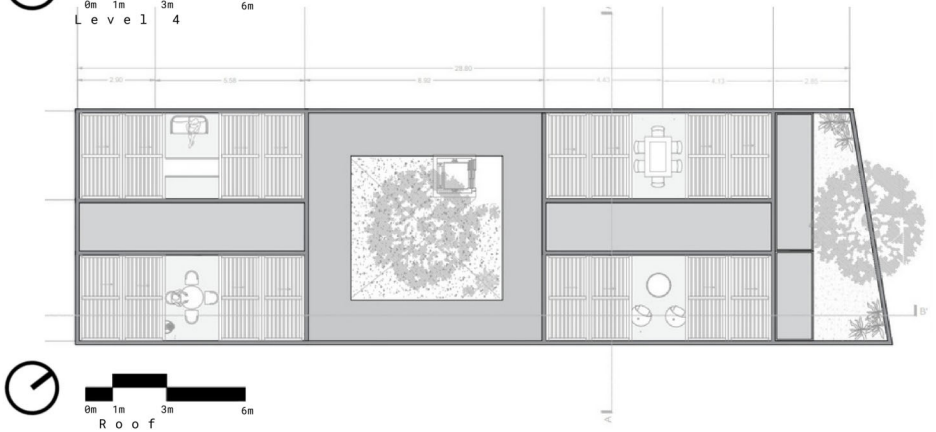
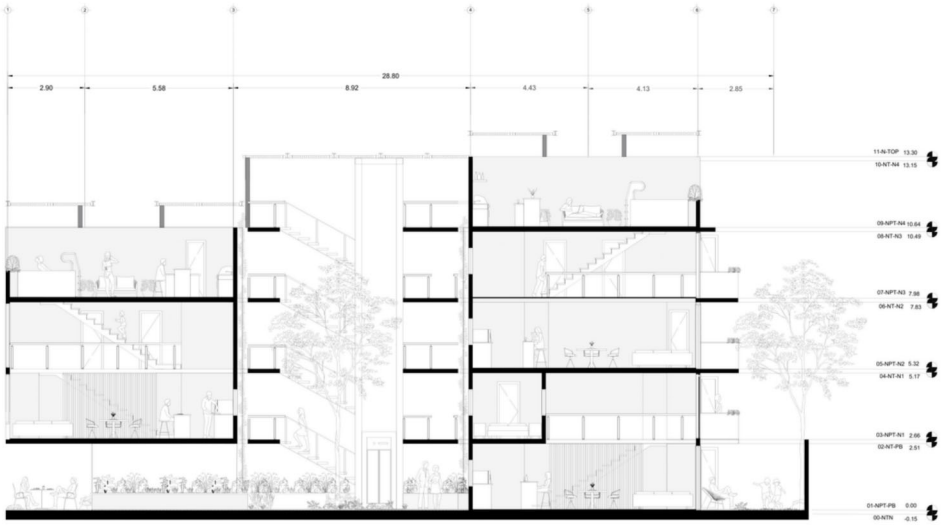
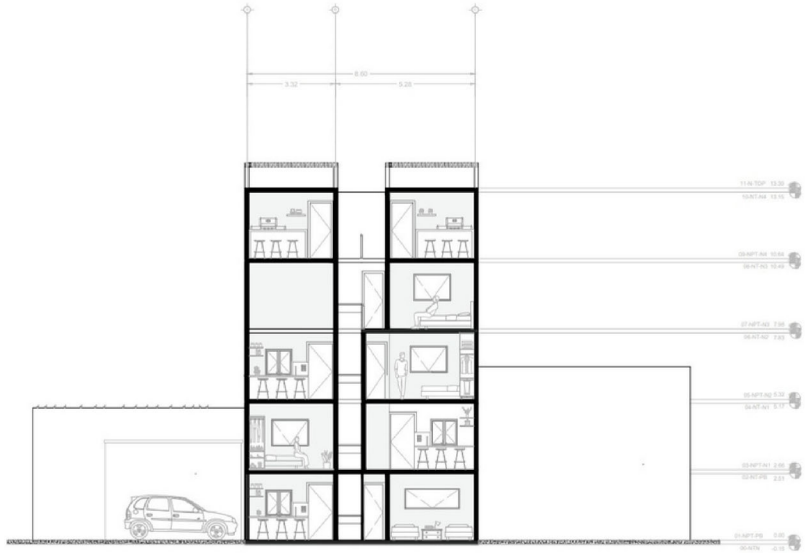


Roof



0m 1m 3m 6m

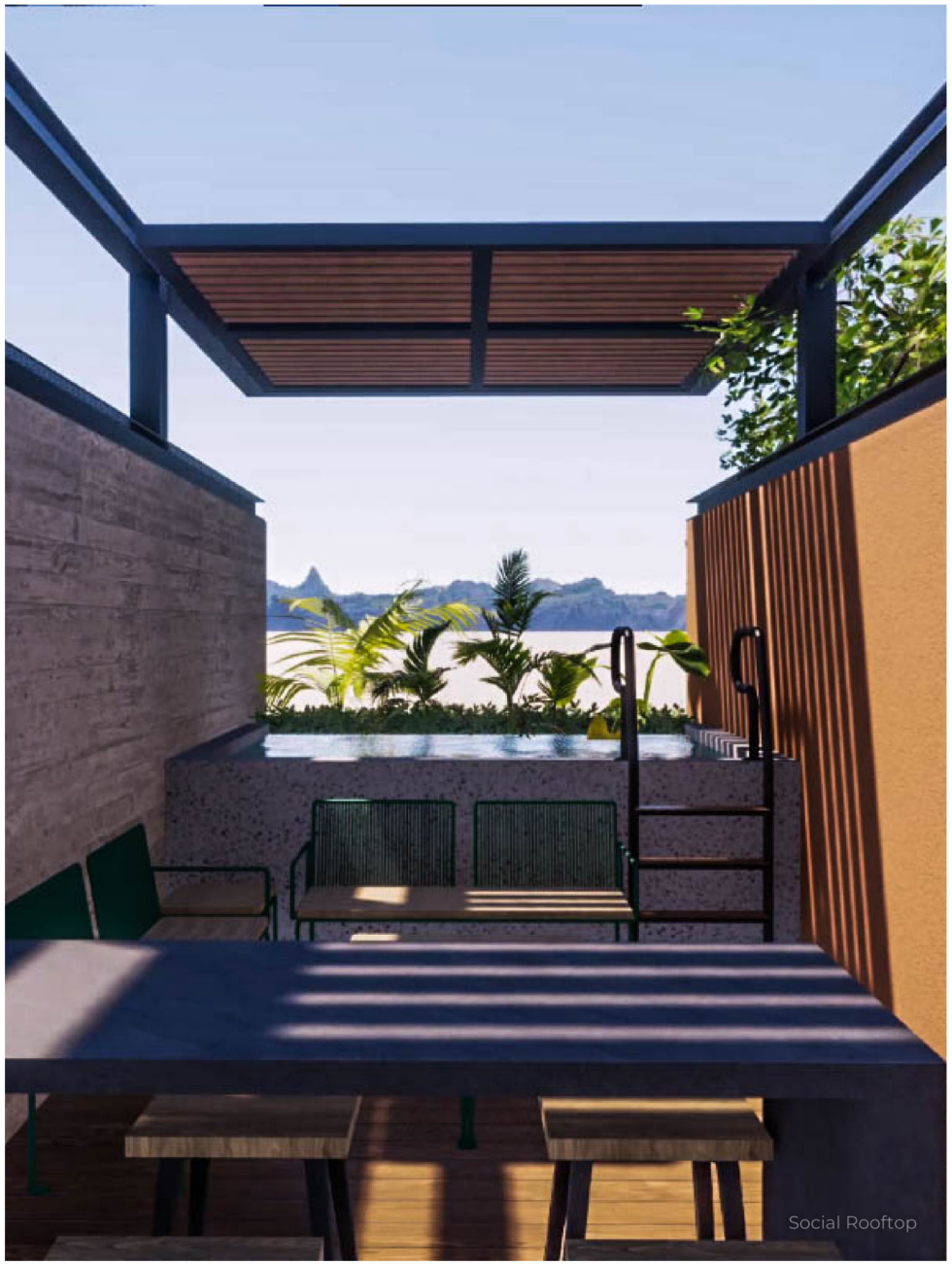
Longitudinal Section



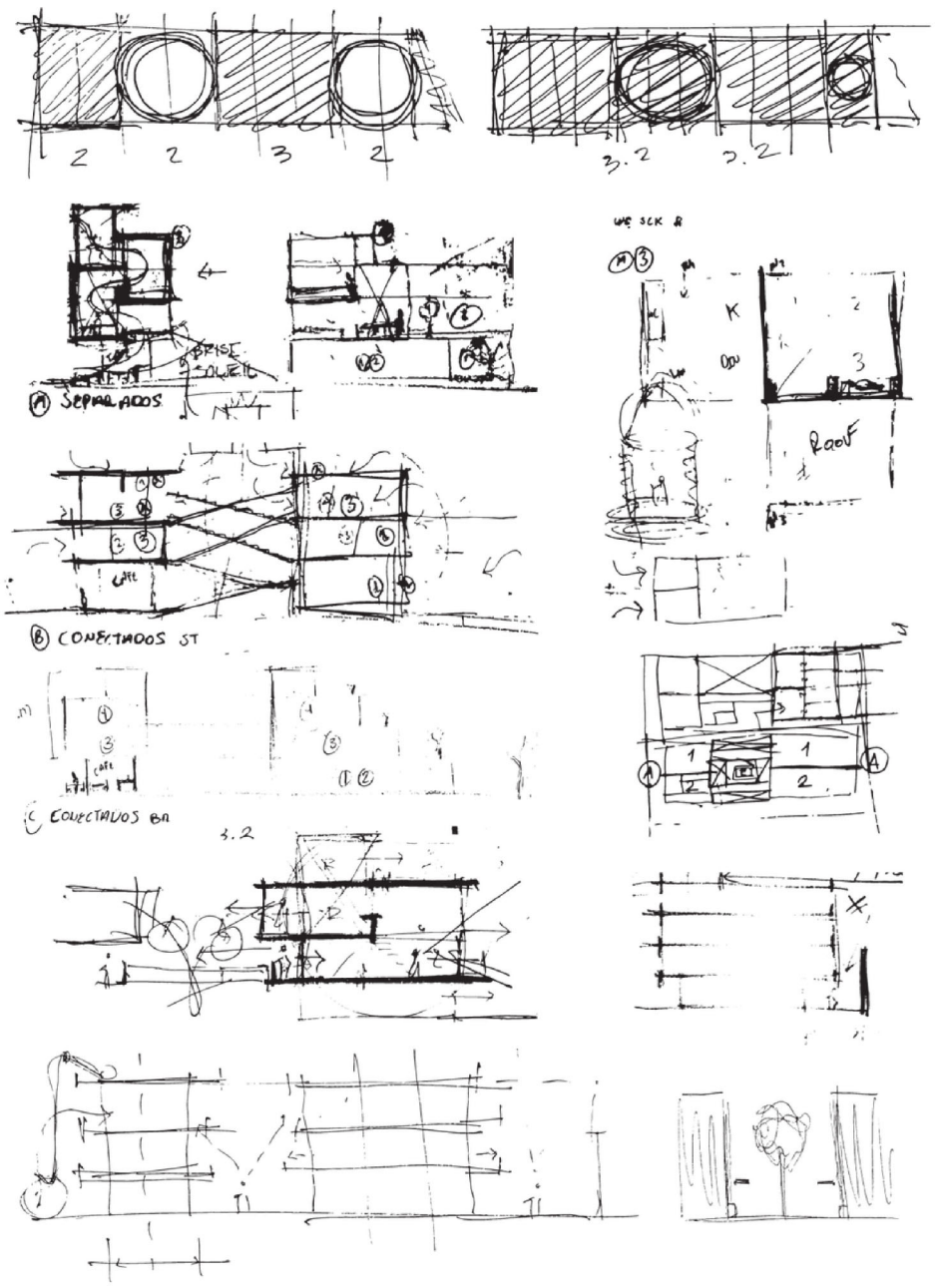
0m 1m 3m 6m

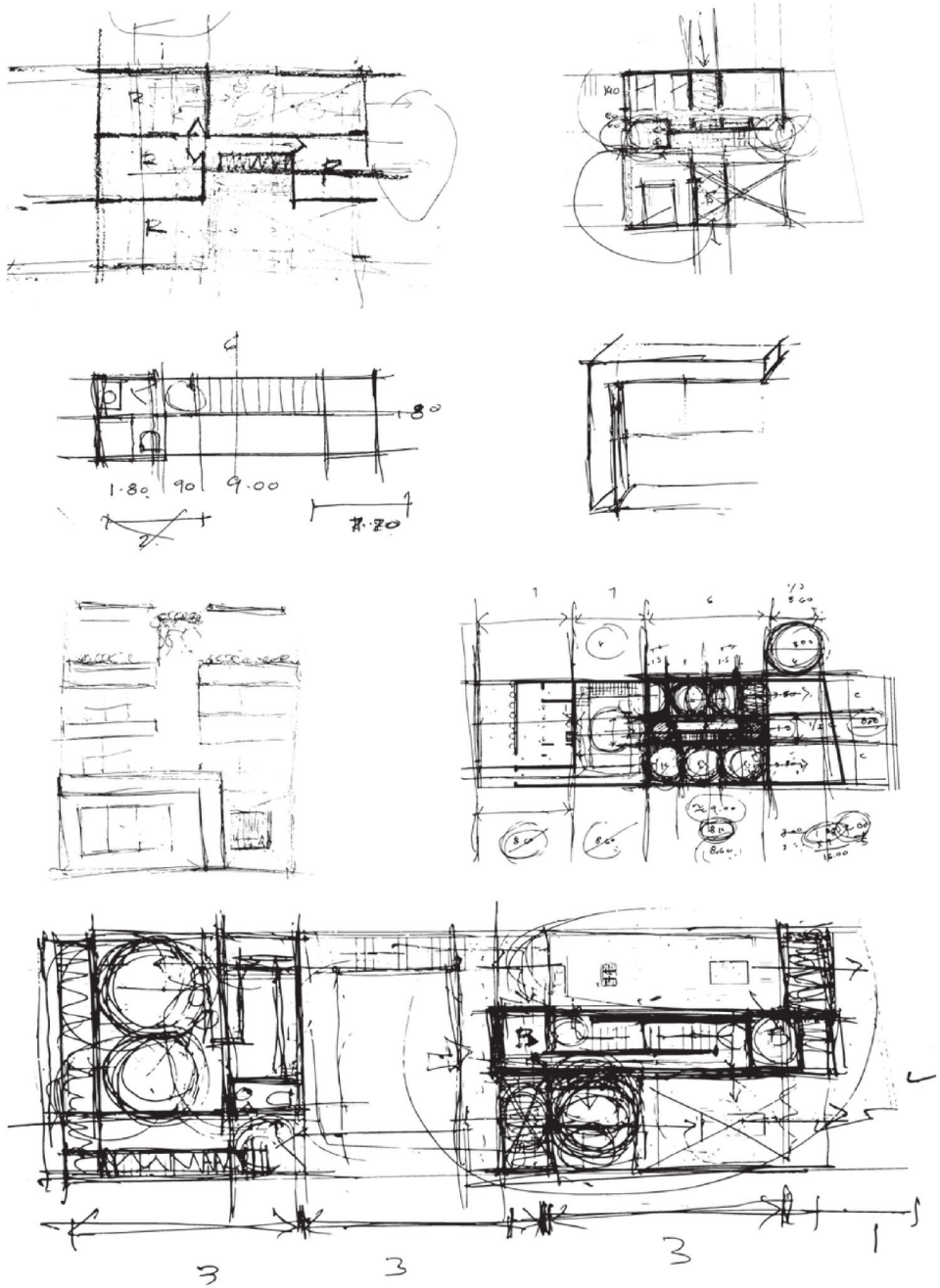
Cross Section

Duplex


Social Rooftop

Distribution

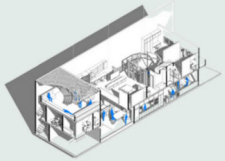

SEPARADOS
BRISE SOLEIL
CONECTADOS ST
CONECTADOS BA
ROOF


ROOF

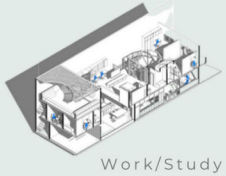
Independencia 221, Casco Urbano, 66200 San Pedro Garza García, N.L.

OCULUS

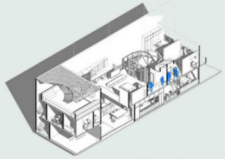
Use Patterns



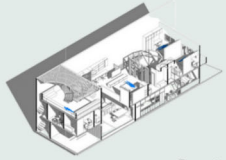
Social



Work/Study



Hygiene



Rest

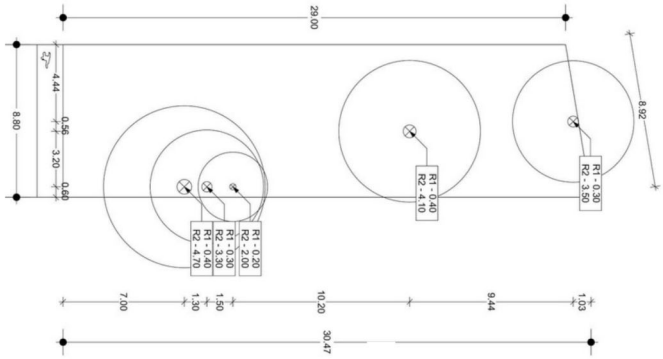
Colaborators:
Edith García
Jarol Mendoza

2023

Oculus

"Design, Nature and Passion."

Oculus, an elegant single-family home designed to meet the unique needs of a modern family in the heart of the Urban Center of San Pedro Garza García. In this exciting project, functionality and nature merge, creating a space that reflects the clients' lifestyle and interests in an exceptional way. The design is addressed for a family made up of a couple, a teenage son and a little girl. The man, a cooking enthusiast, and the woman, an architect, were looking for a home that embodied the harmony between nature and design. The son, a talented drummer, also put forward his own creative requirements. These aspirations, along with a shared love of nature, shaped “Oculus.”

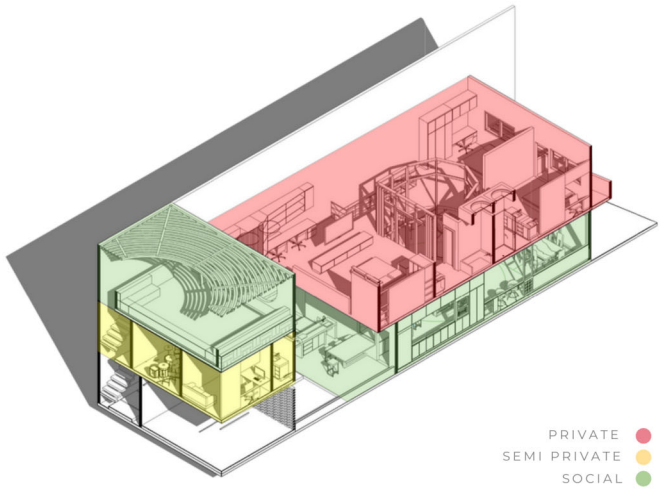


A non-negotiable commitment was the preservation of the existing trees on the land. This fundamental requirement shaped the entire design process, skillfully fusing natural elements with the structure. Each space in "Oculus" offers views of the surrounding greenery, making nature an integral component of the family's daily life.

The design is structured into three distinct but cohesive areas: semi-private, social and private. Each one adapts perfectly to the needs and preferences of the family members. The semi-private area, with exterior access, houses the architect's office and the son's music room. This ensures a harmonious separation between work and family life, while allowing independent entry and exit of clients.

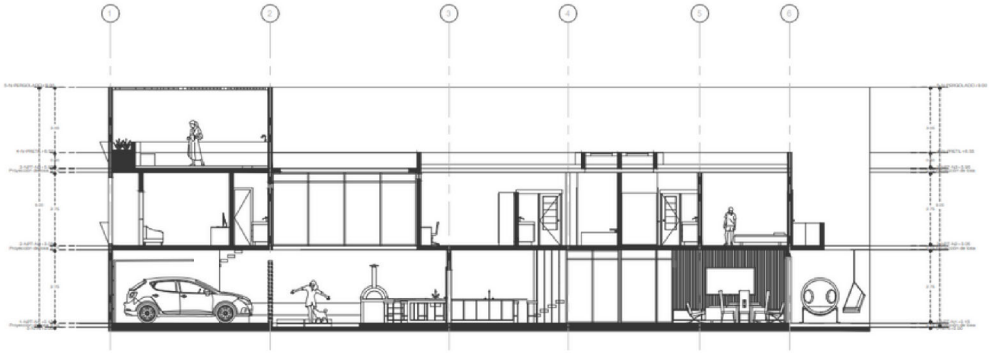
The private area, located on a single level, offers individual spaces designed for the well-being of each family member. For the father, who loves cooking, an integrated kitchen that connects with small harvesting spaces and a pizza oven. The son finds his sanctuary in a music room specially designed to accommodate his drums, encouraging his creativity.

The social area, the heart of the home, is divided into two complementary parts. A rooftop with exterior access that offers views of the city, providing an intimate space to relax. The other part is integrated into the main volume, housing the living room, dining room and kitchen. Here, the kitchen becomes a connection center, uniting indoor and outdoor spaces with connected table-bars, while the garden reflects the passion for nature.

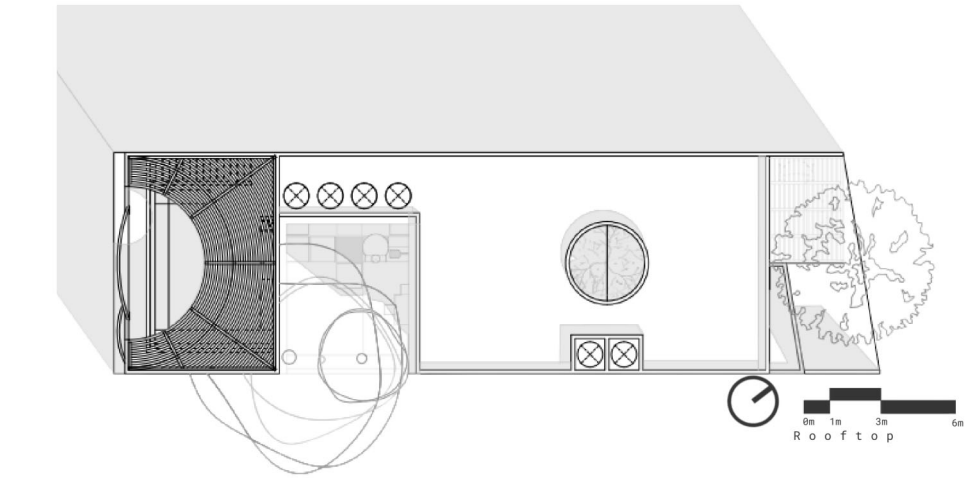


Zoning

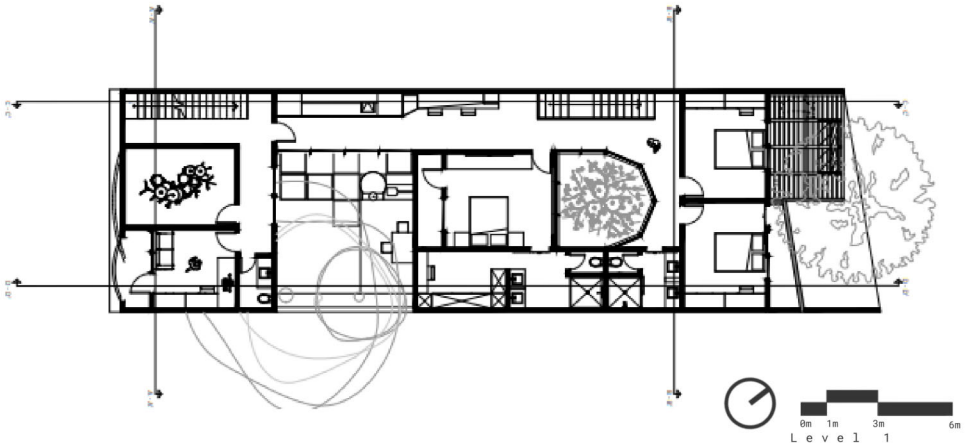
The "oculus" are the distinctive signature of this design, not only as strategically placed light entrances, but also as architectural elements on the façade that play with light and shadows. These key details bring a sense of mystery and movement to the design, adding unique character to the home.



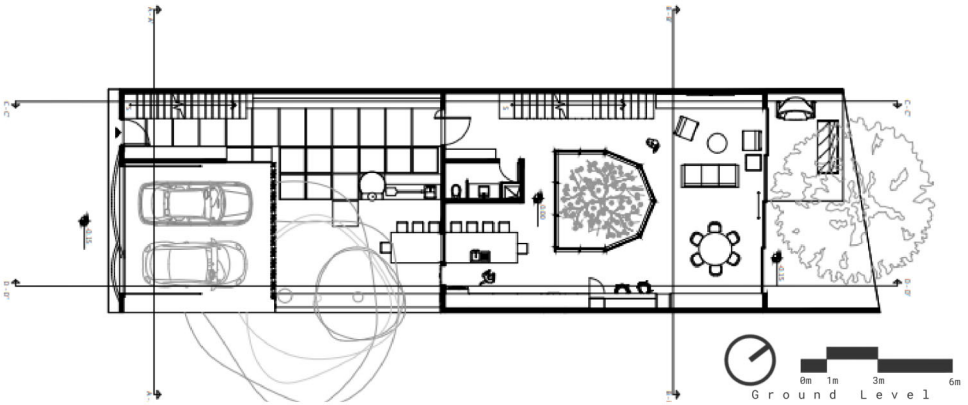
Longitudinal Section



Rooftop



Level 1



Ground Level

Main Facade

Connected Indoor-Outdoor Kitchen

Harvesting Area

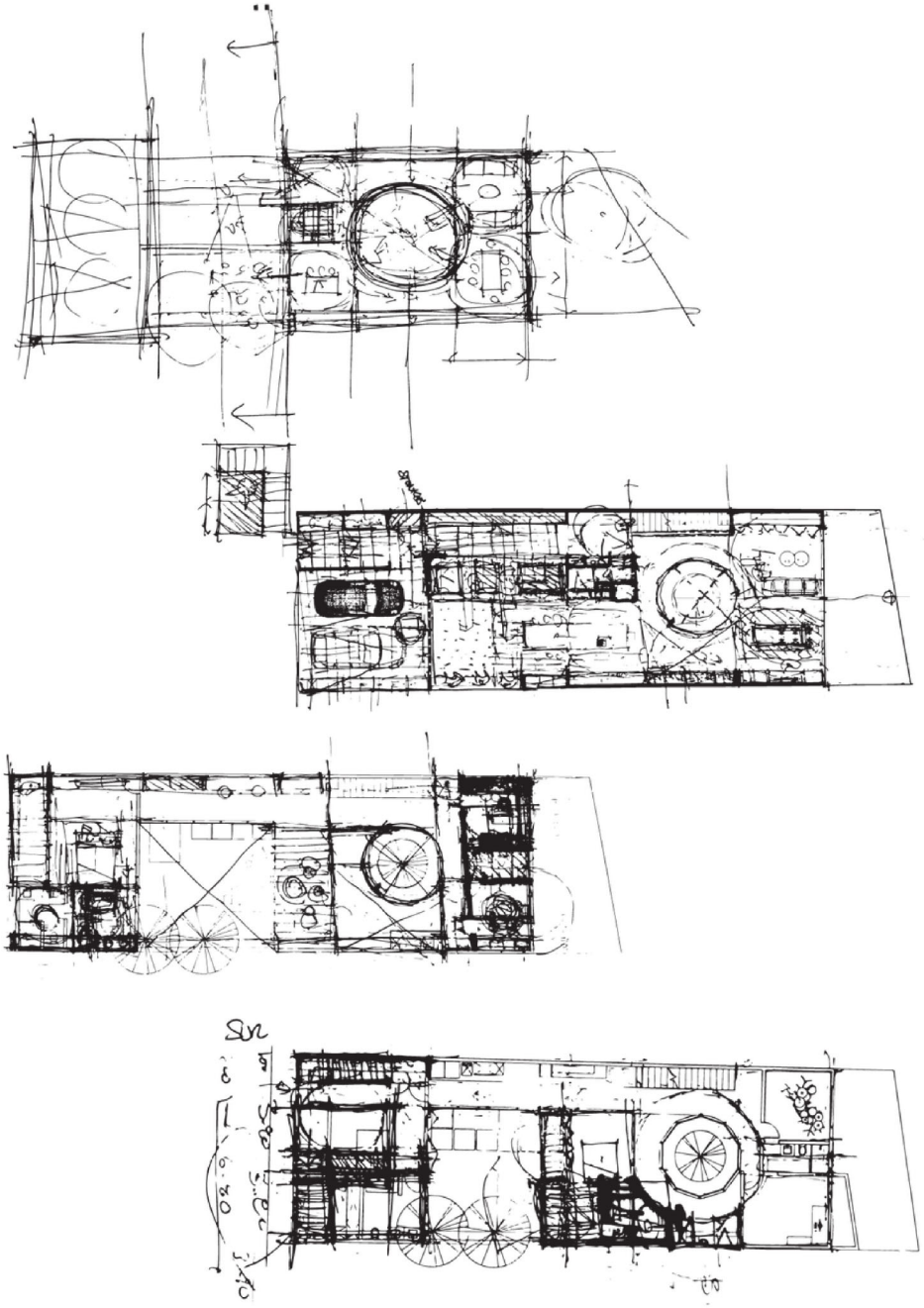
Master Bedroom

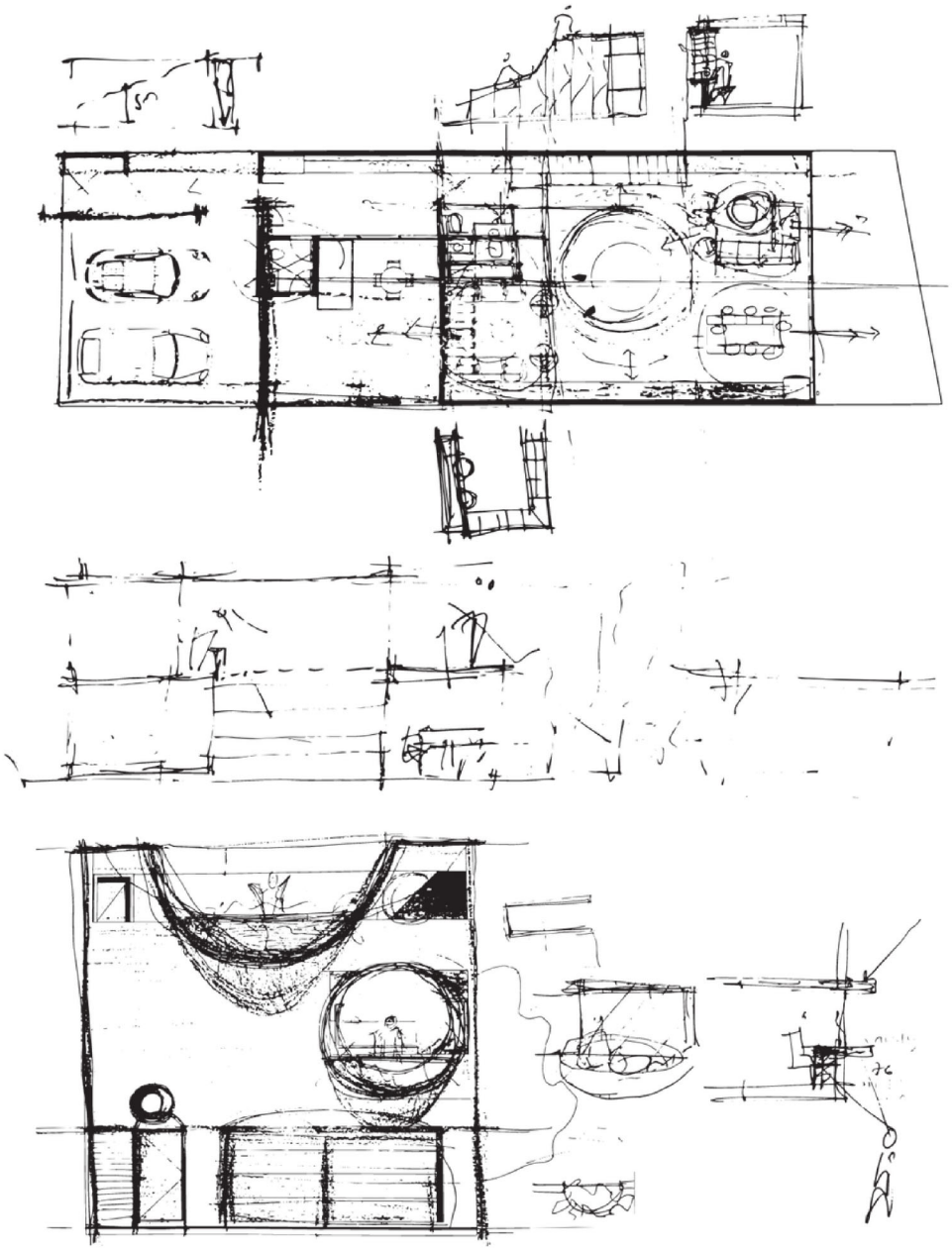
Living Room

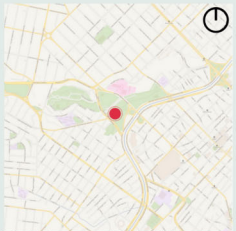
Backyard

Social Rooftop

Connecting Hall



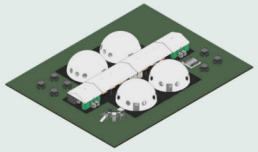




El Palomar, 31203 Chihuahua, Chih.

DOME SHELTER

Ephemeral shelters play a crucial role in responding to natural disasters and emergencies. They provide temporary shelter for affected people, offering security, shelter and comfort at critical times. Additionally, these shelters can be a means to provide medical care, food, and psychological assistance to those in need. Their design must be versatile and adaptable to different situations, which makes them an architectural solution with high social impact.



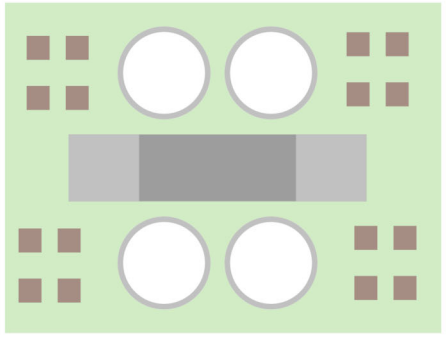
2022

Dome Shelter

"Quick Shelter for Critical Situations"

This project seeks to address a crucial problem in the city of Chihuahua: the lack of preparation to face the heavy rains and their consequent floods, which seriously affect the local population. This ephemeral shelter proposed in Palomar Park, in the neighborhood of the same name south of the city, aims to provide a safe and comfortable shelter for people affected by these natural disasters. This project not only responds to an immediate need, but also focuses on innovation and sustainability in its design and construction.

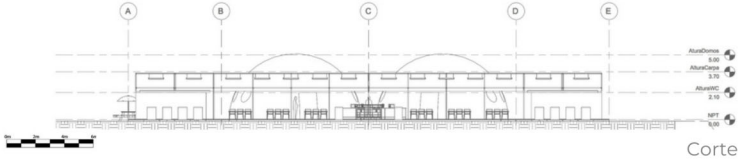
Floods in Chuhuahua



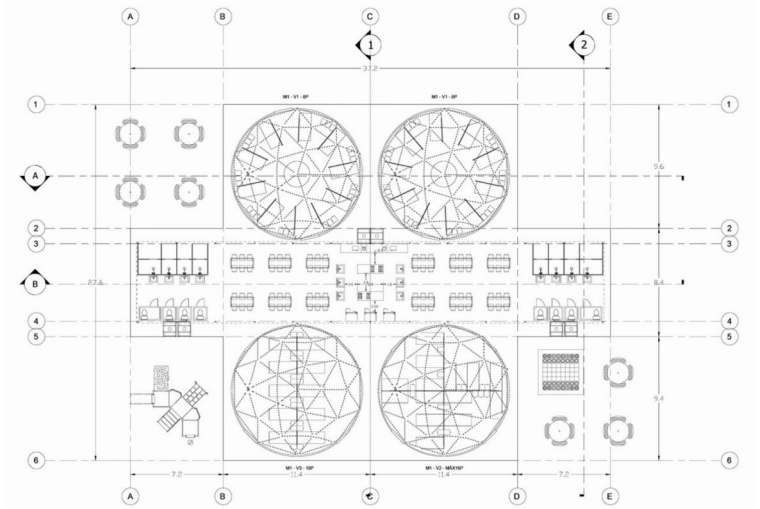
Palomar Park, selected as the location for this project, is a strategic choice due to its altitude and size. Being at an elevated level, the park significantly reduces the risk of flooding. Furthermore, its scale allows for effective implementation and the possibility of creating an architectural structure that integrates with the natural environment.

The ephemeral shelter is made up of five fundamental zones, each designed to provide functionality and comfort: a common kitchen, dining rooms, bedrooms, bathrooms and playgrounds for common areas. This multifunctional design ensures that the basic needs of affected people are met in a welcoming and safe environment.

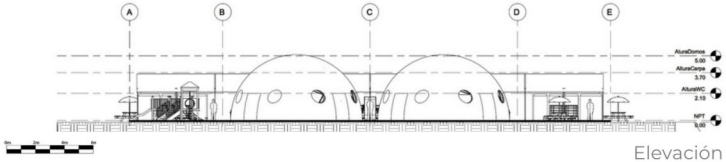
Los domos geodésicos se utilizan en este proyecto debido a sus propiedades únicas que se alinean con el objetivo de proporcionar confort y seguridad. Estos domos ofrecen una distribución uniforme de la temperatura, garantizando condiciones agradables tanto en verano como en invierno. Para garantizar un aislamiento efectivo, se propone el uso del material "efte", una elección que se basa en su capacidad aislante y resistente. El uso de tubería PVC para la estructura de los domos, cubierta de membrana efte, paneles HPL para divisiones y pallets industriales reciclados para los pisos refleja un compromiso con la sostenibilidad y la reutilización de recursos. Estos materiales no solo son duraderos, sino que también reducen la huella ambiental del proyecto, alineándose con los principios de diseño responsable.



Corte B



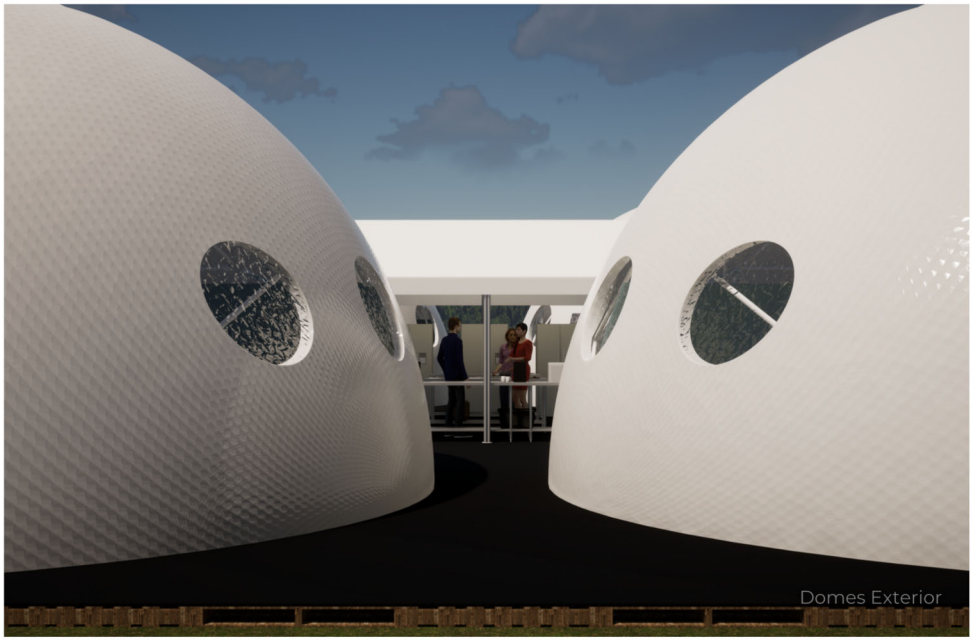
Planta



Elevación Sur

El albergue efímero propuesto en el Parque Palomar en Chihuahua no solo responde a la problemática de inundaciones en la ciudad, sino que también demuestra cómo la arquitectura puede ser una herramienta poderosa para enfrentar desafíos sociales y ambientales. Mediante la integración de innovación, funcionalidad y sostenibilidad, este proyecto se convierte en un ejemplo inspirador de cómo la arquitectura puede marcar una diferencia significativa en la vida de las personas en momentos de necesidad extrema.

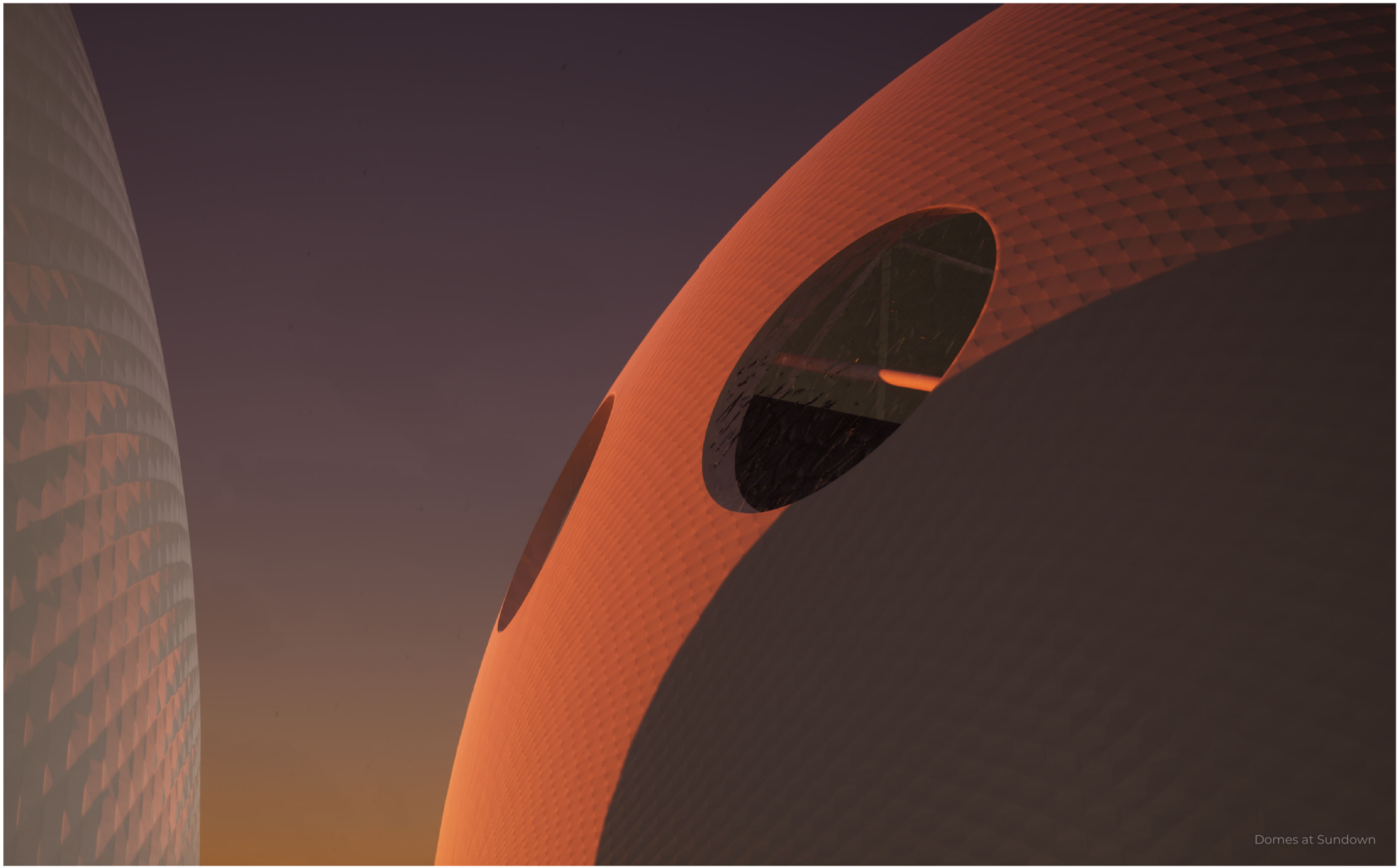
View To Dining Area


Domes Exterior

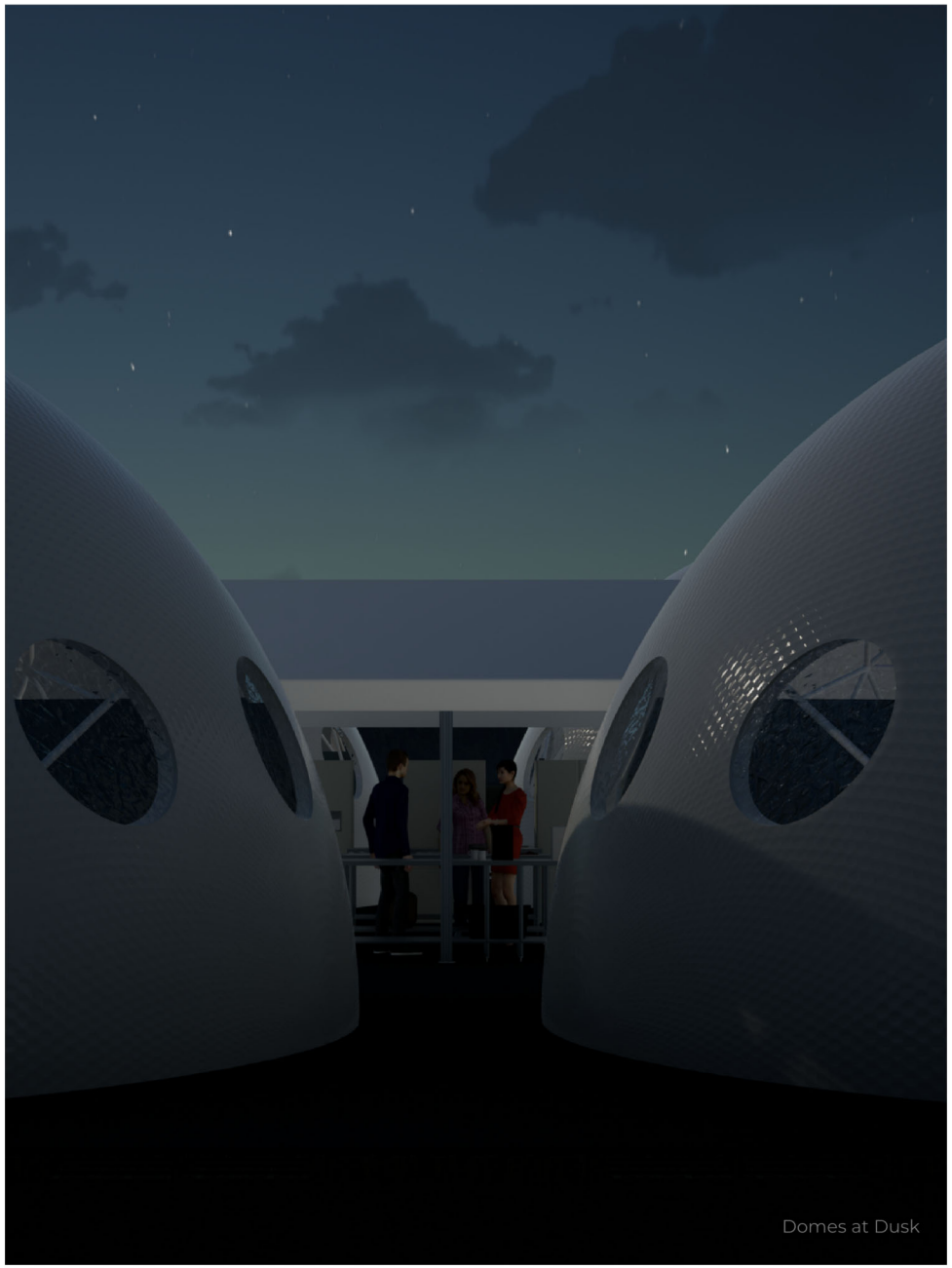
Social Common Areas

Dining Area

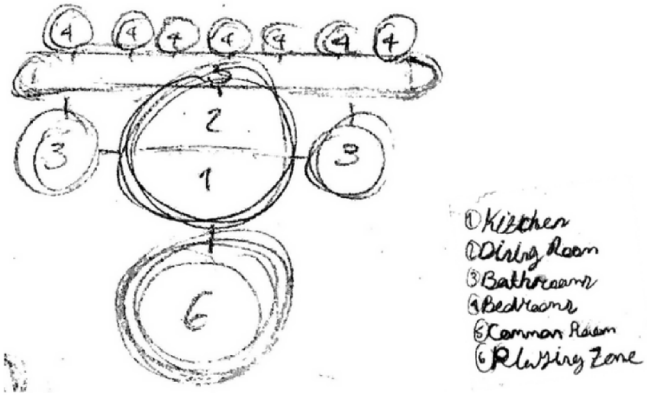
Dome Interior

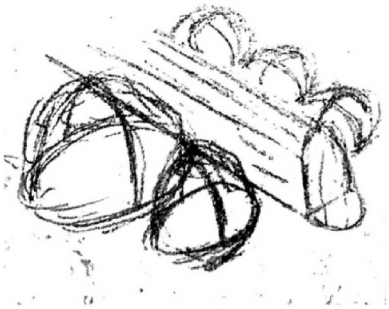


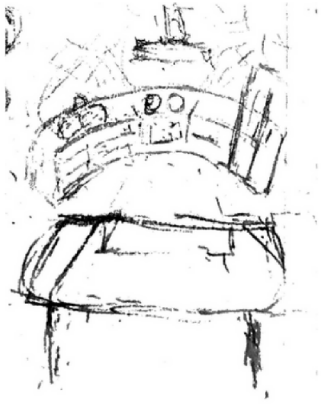
Domes at Sundown



Domes at Dusk



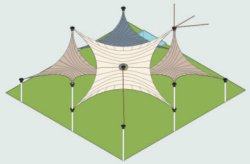




Sin Nombre de Col 18, Saltillo, Coah.

TENSILE SHADE

The design is based on the creation of a composition of sails that follow the shape of a hyperbolic paraboloid, taking advantage of the beauty of its curves and lines. The incorporation of strategically placed masts allows for a visually striking appearance while maintaining the essential functionality of the project.

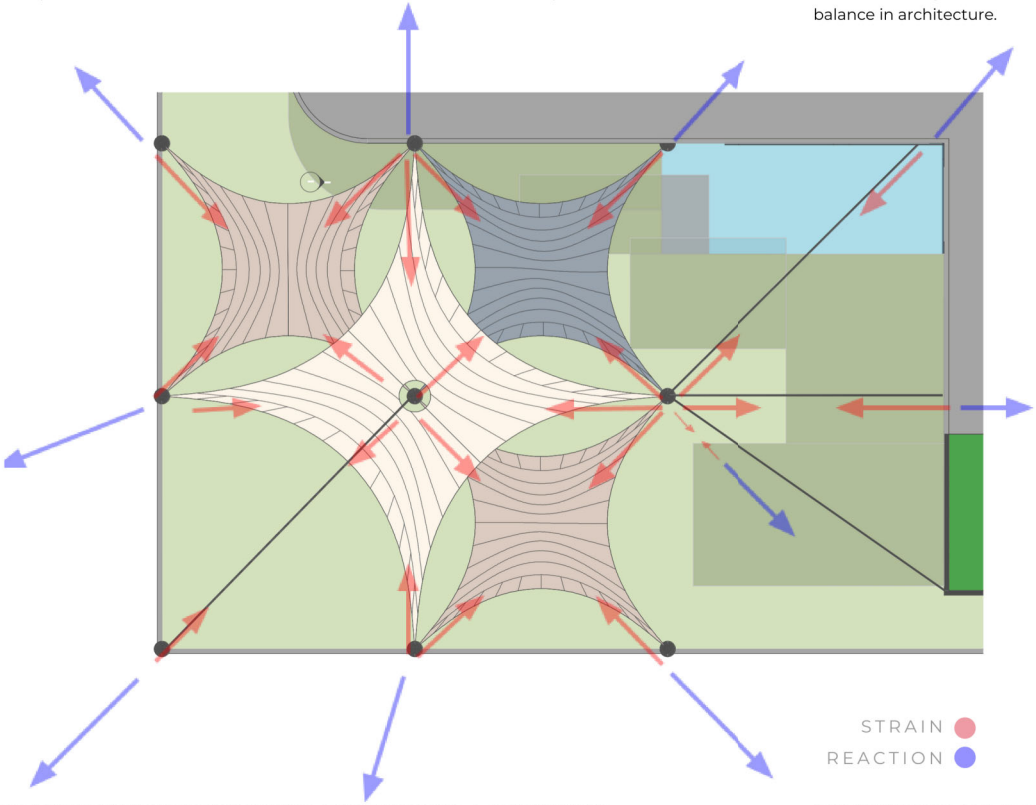


2022

Tensile Shade

"Aesthetics in Balance"

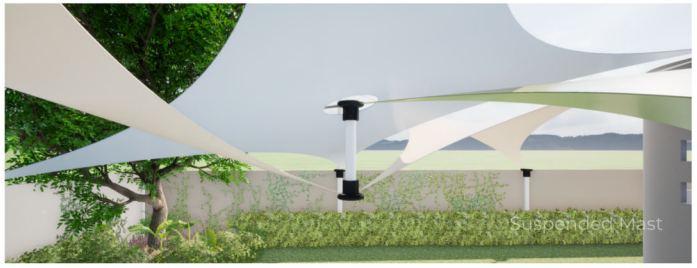
This project presents an innovative and aesthetically attractive solution to provide shade and beautify the garden of a house in Saltillo, Coahuila. The proposal consists of a tensile structure that not only fulfills its practical function, but also adds a distinctive and captivating design to the environment.
To ensure the stability and balance of the structure, some statics concepts were applied. This approach ensures that the structure can withstand the forces applied by material stress and weather conditions. Balance is achieved when the forces compensate each other, thanks to the interaction of the different components.



The relationship between applied forces and reaction forces is essential for the structure to remain stable. The tension generated by the sails and support cables counteracts the forces of gravity and other possible external factors. This ensures that the structure remains in balance and serves its purpose efficiently.
The integration of the different elements, from the masts to the sails and anchors, results in an architectural structure that is aesthetically pleasing and sustainable. Combining the beauty of the hyperbolic paraboloid with the practical functionality of providing shade creates a unique and attractive space.

The relationship between the forces, vectors and elements of the sails is essential to understand the result of the structure. The sail membrane and tension cables work together to generate resultant forces that counteract external forces, maintaining proper tension in the structure. The masts, for their part, play a crucial role in providing the anchorage necessary to keep the fabric taut and prevent it from flexing inward.

An outstanding feature of this project is the central mast suspended in the air. Thanks to a clever design that capitalizes on the forces of the sails and cables, this mast appears to defy gravity by staying in position without visible supports. This feature not only adds an element of wonder to the design, but also demonstrates a deep understanding of forces and balance in architecture.



This architectural project represents an impressive fusion between design and functionality, demonstrating how deep knowledge of forces, balance and structural physics can lead to the creation of masterpieces that enhance both the environment and the user experience. The hyperbolic paraboloid sail in Saltillo, Coahuila, not only provides shade, but also tells a story of innovation, engineering and creativity in the world of architecture.

View from Deck

Side View

Poolside View

General Side View

Special Thanks to Mentors:

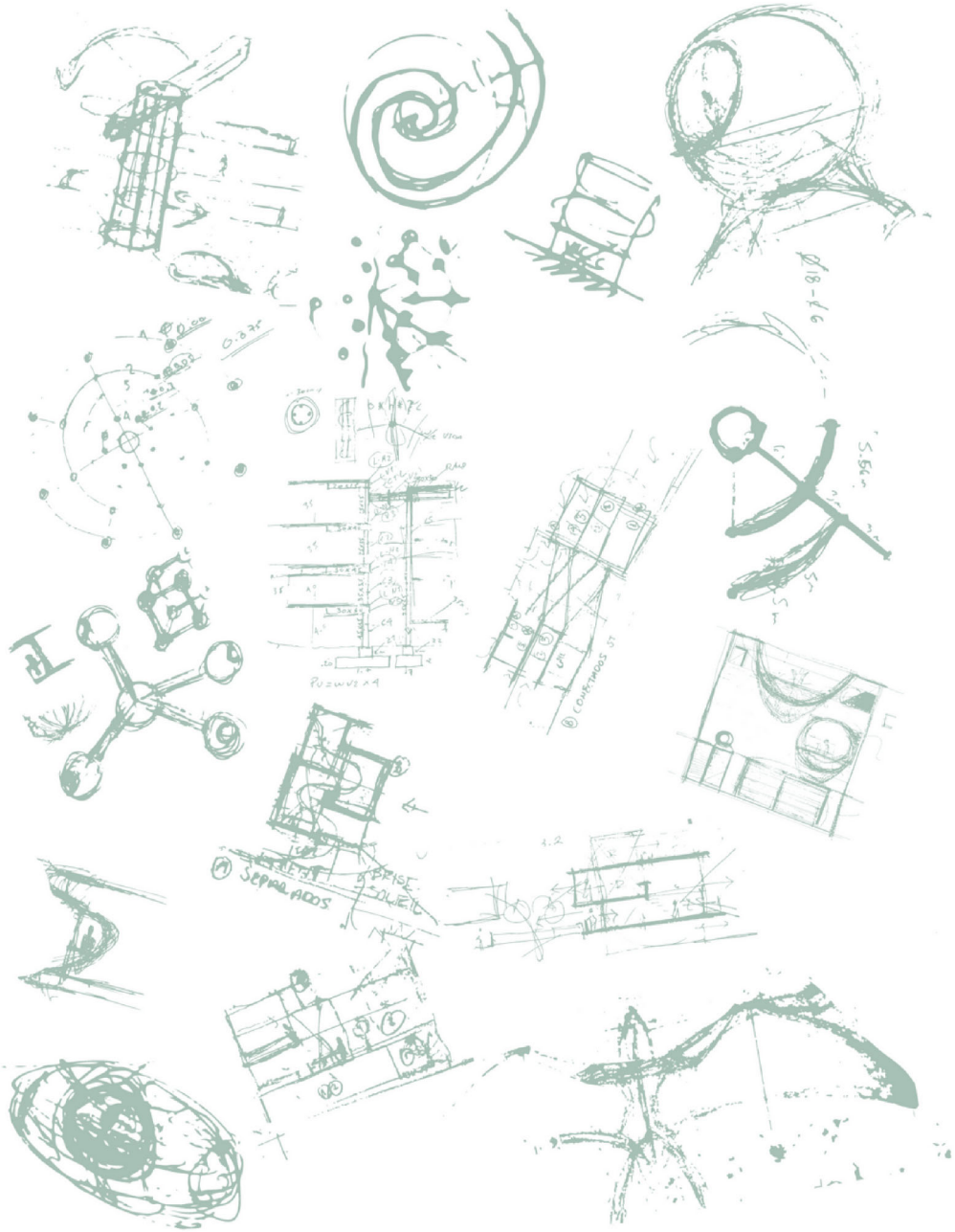
Arq. Oscar Bulnes V

Dr. Pedro Pacheco V

Arq. Juan Carlos López A

Arq. Roberto García R

Arq. Ivette Cavazos E



"Our reactions are shaped by the space we inhabit."